Supporting Homesickness and Loneliness in International Students:

Designing a Wellbeing-Maintenance Calling Tool

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Abstract

International students experience documented wellbeing pressures linked to homesickness and loneliness, while sustained contact across distance is often complicated by academic workload, time-zone differences, and the effort of initiating meaningful calls. Existing communication tools enable calling but provide no support for maintaining contact over time. This dissertation reports the design and evaluation of *keepclose*, a high-fidelity Figma prototype designed as a wellbeing-oriented maintenance layer over the calling tools international students already use. The project tests the hypothesis that a low-friction, calm, autonomy-respecting calling layer — one that surfaces drift without instrumenting relationships, reduces the friction of acting on intent, and refuses the gamified vocabulary of productivity and habit-tracking apps — can plausibly support the wellbeing-oriented contact maintenance international students need. The project followed a human-centred design process informed by Peters' (2023) wellbeing-supportive design framework. A 50-respondent survey of international students informed a structured set of functional and non-functional requirements, which shaped a seven-screen prototype supporting three core user journeys. The prototype was evaluated through scenario-based walkthroughs with ten international students. Task-completion data showed that the interaction model carried all participants through the three core journeys without outright failures. Qualitative findings provided qualified support for the hypothesis at the level of first-contact plausibility, with participants responding most positively to the deliberate absence of streaks, scores, and gamified comparison. Refinements were identified for microcopy, label editing, and communicating the system's protective principles more clearly. The project contributes to the design space of wellbeing-supportive relational technology by providing an empirically grounded example of *deliberate restraint* as a design principle and identifying directions for future longitudinal evaluation.

Declaration

"I declare that this dissertation represents my own work, except where otherwise stated."

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The interactive Figma prototype is available at:
https://www.figma.com/proto/0B8tftOfGUGLy5E3CEovnf/CSC3094--Prototype?node-id=5-2&p=f&t=t9PPbauTl4bC6bGU-1&scaling=scale-down&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=5%3A2

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Chapter 1: Introduction

1.1 Motivation

Homesickness and loneliness are well-documented features of the international student experience *(Zheng et al., 2023).* Students moving abroad are simultaneously separated from familiar home networks and placed in environments where local social ties have not yet formed, and the resulting experience can be psychologically distressing as well as practically difficult *(Zheng et al., 2023)*. Although students routinely use digital communication technologies to sustain contact with home *(Kelly et al., 2021)*, everyday pressures — academic workload, time-zone differences, and the simple effort of initiating a meaningful call — make this harder than it sounds. Calls that are intended drift unattended into the background, and what begins as a postponement becomes a gap. Existing communication tools enable calling but do not support its maintenance over time: they assume the user already knows when to call, who to call, and at what frequency. For international students operating under sustained academic and life pressures, this assumption does not hold. The opportunity the project addresses is the gap between the communication tools students already rely on and the wellbeing-oriented support that a *maintenance layer* over those tools could plausibly provide.

1.2 Aim and Hypothesis

The project's central aim was to design and evaluate a high-fidelity mobile app prototype that supports international students in maintaining meaningful voice and video calls with the people who matter to them. The prototype, named *keepclose*, is positioned as a wellbeing-oriented *maintenance layer* over the existing calling tools students already use, rather than as a replacement for them. Its design tests a specific hypothesis: that a low-friction, calm, autonomy-respecting calling layer — one that surfaces drift in contact frequency without instrumenting relationships, reduces the friction of acting on intent, and refuses the gamified vocabulary of productivity and habit-tracking apps — can plausibly support the wellbeing-oriented contact maintenance international students need. This hypothesis is informed by the wellbeing-supportive design framework articulated by Peters (2023), which provides the theoretical lens through which the project's design decisions were framed. The evaluation reported in Chapter 4 produces qualified support for the hypothesis at the level of first-contact plausibility; whether the design supports wellbeing in sustained real-world use is identified as future work (Chapter 5).

1.3 Objectives

The project was structured around five objectives, derived from the proposal and refined as the work progressed:

O1. *Review literature on homesickness and loneliness among international students and on HCI/digital approaches that support wellbeing maintenance and social connection, deriving initial design directions.* Addressed in Chapter 2.

O2. *Conduct user research with international students to understand factors contributing to homesickness and loneliness, patterns of connection drift, time-zone challenges, and*

preferences around reminders and reflection on contact routines. Addressed in §3.2 through a 50-respondent online survey.

O3. *Translate the literature and user research findings into a structured set of design requirements and a prototype information architecture covering Home and Local Circles, contact frequency settings, time-zone-aware contact suggestions, call initiation via external platforms, and outcome logging.* Addressed in §3.3 (requirements) and §3.4 (information architecture).

O4. *Build a high-fidelity Figma prototype implementing the core end-to-end journeys and key screens.* Addressed in §3.5 through a seven-screen prototype with three wired user journeys.

O5. *Evaluate the prototype with international students using scenario-based walkthroughs and structured feedback, then document findings and limitations.* Addressed in §3.6 (evaluation methodology) and Chapter 4 (findings).

One change to the proposal's plan is worth noting. The proposal had described a post-evaluation iteration cycle following the initial evaluation; this was scoped as optional during the project, and the time was directed into the analysis of the single evaluation round rather than into a partially completed second iteration. The implications are revisited as future work in Chapter 5.

1.4 Approach in brief

The project followed a human-centred design process *(Sharp, Rogers and Preece, 2019)*, in which the design responds to evidence about people and their practices rather than to designer assumptions. Within this framing, the design was anchored in the wellbeing-supportive design perspective articulated by Peters (2023), introduced in Chapter 2. The empirical work proceeded in two phases: a 50-respondent online survey of international students, which informed the requirements and design direction (§3.2, §3.3); and a ten-participant evaluation of the prototype with international students using scenario-based walkthroughs (§3.6, Chapter 4). The deliverable is a high-fidelity Figma prototype consisting of seven screens supporting three end-to-end user journeys; the rationale for the prototype-as-deliverable scoping is recorded in §3.1.

1.5 Dissertation structure

The remainder of the dissertation is organised into four chapters. Chapter 2 (Background Review) reviews the literature on homesickness and loneliness in international students, on digital technologies for sustaining connection across distance, and on the broader HCI tradition of designing for awareness and lightweight social presence; it also surveys existing applications in adjacent product categories and identifies the gap *keepclose* is designed to occupy. Chapter 3 (What Was Done and How) describes the user research, requirements, information architecture, prototype design, and evaluation methodology — the full sequence of project activities through which the prototype was developed and evaluated, presented alongside the evidence that shaped each decision. Chapter 4 (Results and Evaluation) reports the findings from the ten-participant evaluation across task-completion data, seven thematic findings from the open-ended responses, and a requirements-by-evidence assessment, before discussing what the findings collectively imply for the project's design hypothesis and the limitations of the evaluation method. Chapter 5 (Conclusions) assesses the extent to which

the project's objectives were met, summarises what was learned about both the design and the methods used, and identifies directions for future work.

Chapter 2: Background Review

2.1 Chapter overview

This chapter establishes the conceptual and technological context within which the *keepclose* prototype was designed and evaluated. The literature review (§2.2) covers homesickness and loneliness among international students, how digital technologies are used to sustain connection across distance, the broader HCI tradition of designing for awareness and lightweight social presence, and the design lens — wellbeing-supportive design — through which the project's interaction decisions were framed. The state-of-the-art review (§2.3) examines existing applications in adjacent product categories, identifying what each contributes to or contradicts in the design direction the project takes. The chapter closes (§2.4) by identifying the gap *keepclose* is designed to address, handing off to the design work described in Chapter 3.

2.2 Literature review

2.2.1 Homesickness and loneliness in international students

Homesickness and loneliness are well-documented features of the international student experience *(Zheng et al., 2023)*. Although the constructs overlap, they are distinct: homesickness is a longing for familiar people, places, and routines associated with home *(Stroebe, Schut and Nauta, 2015)*; loneliness is the subjective experience of unwanted social disconnection *(Perlman and Peplau, 1981)*. For international students the two frequently co-occur, because the move abroad removes everyday contact with familiar home networks while placing students in an environment where local social ties have not yet formed.

Empirical work has consistently identified loneliness as common rather than exceptional in the international study experience. Sawir *et al.* (2008), interviewing 200 international students in Australia, reported that two-thirds described experiences consistent with loneliness in two distinct forms — personal loneliness from separation from family and friends, and social loneliness from the absence of an established cultural community in the host country. Similar patterns have been reported in the United States *(Bhochhibhoya et al., 2018)* and across multiple host countries *(Hendrickson, Rosen and Aune, 2011)*, with friendship-network composition — particularly the balance of co-national and host-national friendships — significantly predicting life satisfaction.

More recent qualitative work has deepened understanding of these experiences. Zheng *et al.* (2023), drawing on qualitative survey data from the BBC Loneliness Experiment, identified several recurring features of how international students describe loneliness. The experience can be psychologically distressing rather than merely uncomfortable; it is shaped by factors particular to the international-student context — separation from home networks, cultural and linguistic adjustment, and the loss of small everyday interactions — and it can be self-reinforcing, as disconnected students withdraw further. Stigma compounds this: reluctance to disclose loneliness, partly because study abroad is framed as a positive opportunity. These findings extend earlier work by Brunsting, Zachry and Takeuchi (2018), whose review identified social support as a consistent protective factor against the psychological distress of the international transition.

Two features of this body of work matter for the design direction the project adopts. First, the literature treats international student loneliness as a *normal* response to a difficult situation rather than a clinical condition, framing the appropriate response as supportive rather than therapeutic *(Sawir et al., 2008; Zheng et al., 2023)*. This is consistent with the project's wellbeing-maintenance approach, which helps students sustain ordinary coping practices rather than delivering any clinical intervention. Second, because the experience is shaped partly by reduced contact with familiar people, sustaining contact with home networks is plausibly one of the everyday practices through which wellbeing is maintained — a premise the user research directly tests (Chapter 3), supported by HCI work on how students use digital technologies for this purpose *(Kelly* et al.*, 2021, discussed in §2.2.2)*.

2.2.2 Digital technologies for staying connected across distance

International students rely heavily on digital communication technologies to sustain relationships with family, friends, and home networks while abroad. Madianou and Miller (2013) introduced *polymedia* to describe how users move fluently between multiple channels — voice, video, text, asynchronous, synchronous — selecting whichever fits the emotional and practical character of a particular interaction. For international students, this polymedia repertoire is the everyday infrastructure through which connection with home is sustained.

The most directly relevant HCI study is Kelly *et al.* (2021), whose interview-based work with university students examined how digital technologies are used to alleviate homesickness. They identified that students use technology not only for direct communication with family and close friends but also for *peripheral* connection — watching content from home, browsing photos, or maintaining lightweight awareness through social media. Kelly *et al.* frame this as an everyday coping practice rather than an occasional remedy for acute distress, arguing that design for homesickness should support sustainable practices over time rather than crisis-moment interventions. This framing is foundational to the project's wellbeing-maintenance approach.

Voice and video calls hold a particular place in students' coping repertoire because they more closely approximate the texture of in-person contact. Longhurst (2013), researching transnational families using video calling, observed that the visual and synchronous qualities of video calls support a sense of *co-presence* that purely textual communication cannot — being meaningfully *with* someone across distance, rather than merely exchanging messages.

The broader evidence base on digital interventions for loneliness reinforces these findings. Welch *et al.* (2023), in an evidence-and-gap-map review of digital loneliness interventions in older adults, identified technology-enabled social interaction — particularly video calling — as one of the most commonly studied and promising mechanisms. Although the population is older adults rather than international students, the underlying mechanism is analogous, supporting the broader plausibility of calling-based interventions for connection-related wellbeing.

Together, this work establishes a layered foundation for the design hypothesis. Digital communication is already the *de facto* infrastructure international students use to sustain home relationships; voice and video calling have particular value within it because they support co-presence; and the design challenge is not to introduce new communication tools but to support more sustainable use of the tools students already rely on. This framing —

design as a layer above existing communication infrastructure rather than a replacement — directly informs the architectural decisions in Chapter 3.

2.2.3 Connection, awareness, and lightweight presence in HCI

Beyond the international-student literature, a broader strand of HCI research addresses how technology can support a sense of social connection and lightweight presence across distance. This work provides design vocabulary and conceptual framings that the project draws on in adapting them to the international-student context.

The foundational HCI work here is Dourish and Bly (1992) on Portholes, an early system supporting awareness in distributed work groups by sharing low-resolution images of remote colleagues' workspaces throughout the day. The contribution, beyond the specific system, was the introduction of *awareness* as a design target distinct from communication: people benefit from a continuous, lightweight sense of who is around and what they are doing, even in the absence of direct interaction. The underlying insight — that supporting connection can be valuable independent of supporting communication itself — informs the project's design philosophy.

Hassenzahl *et al.* (2012) propose a framework of *strategies for mediating relatedness*, identifying six recurring approaches in HCI design: awareness (knowing what someone is doing), expressivity (sharing thoughts or feelings), physicalness (mediated touch or proximity), gift giving (small acts of giving), joint action (shared activities across distance), and memories (revisiting shared past). Their analysis is grounded in self-determination theory, in which relatedness is one of three psychological needs supporting wellbeing. Two of these strategies are most relevant to *keepclose*: awareness and expressivity. The framework provides language for the design moves the prototype makes — surfacing drift (awareness) and supporting call initiation (expressivity) — and locates these within a broader vocabulary HCI has developed for designing relational technology.

Neustaedter and Greenberg (2012), studying couples sustaining intimate relationships across distance, identified the importance of *always-on* video links and small, low-effort communicative acts in maintaining a sense of shared everyday life. Although the context is romantic rather than family relationships, the insight — that frequency and lightness of contact matter as much as depth — is directly applicable. The project's *low-friction* call layer (NFR5) reflects the same intuition: contact that requires effort is contact that can be deferred indefinitely.

Together, this broader HCI literature provides two contributions to the project. It establishes that *awareness* and *connection* are valid design targets distinct from explicit communication, and it offers theoretical vocabulary — relatedness, awareness, expression, lightweight presence — for describing the design moves *keepclose* makes. The state-of-the-art review in §2.3 examines how these principles play out in existing applications.

2.2.4 Designing for psychological wellbeing

The previous sub-sections establish *what* a tool of this kind might usefully address. This sub-section addresses *how* it should be designed.

The design lens this project adopts is the *wellbeing-supportive design* framework articulated by Peters (2023), which proposes guidelines for designing user experiences that support

psychological wellbeing over time, drawing on self-determination theory and positive-psychology research. Three features matter for the project. First, *autonomy* is treated as foundational: designs that support voluntary engagement and meaningful choice are associated with positive wellbeing outcomes, while those operating through pressure or obligation are associated with the reverse. Second, the framework distinguishes *low-pressure, supportive* design moves from *high-pressure, performance-oriented* ones, identifying the latter as inappropriate for wellbeing-oriented contexts even when effective at short-term behavioural change. Third, the framework attends to *sustained* wellbeing rather than single-interaction satisfaction.

These three features align directly with *keepclose*'s design decisions. The autonomy emphasis informs the rejection of mandatory tracking and automatic data access (NFR3, NFR6); the low-pressure framing informs the rejection of streaks, scores, and gamified mechanisms (NFR2); and the sustained-wellbeing framing informs the emphasis on routine maintenance and conservative reminder design (NFR1, NFR4).

Persuasive design and gamification have produced demonstrable behavioural results in domains such as fitness and language learning *(Hamari, Koivisto and Sarsa, 2014)*. The argument for adopting a wellbeing-supportive lens for *keepclose* is not that these approaches are wrong in general, but that they are inappropriate for the relational-wellbeing domain — where the mechanisms that drive engagement in performance-oriented contexts can produce guilt, pressure, and a reframing of intrinsic care as extrinsic obligation. The user research (§3.2.3) corroborates this, identifying streak mechanics, guilt-inducing wording, and the perception of being monitored as the strongest deterrents among surveyed international students.

The Peters (2023) framework provides both the *vocabulary* the project uses to articulate its non-functional requirements (Chapter 3) and the *justification* for the deliberate omissions that distinguish *keepclose* from neighbouring product categories, examined in §2.3.

2.3 State-of-the-art: existing applications

This section examines existing applications in four product categories adjacent to the project, focusing on what *keepclose* takes from or deliberately departs from each rather than on exhaustive feature comparisons.

2.3.1 General-purpose calling and messaging applications

The infrastructure international students rely on consists of general-purpose calling and messaging applications such as WhatsApp *(Meta Platforms, n.d.)*, FaceTime *(Apple Inc., n.d.)*, Telegram *(Telegram FZ-LLC, n.d.)*, and WeChat *(Tencent, n.d.)*. These have become the universal substrate for transnational personal communication: in the user research, every surveyed student reported using at least one as their primary means of contact with home (§3.2.3).

What these applications do not provide is any layer of support for the *maintenance* of contact over time. They are designed around the transactional unit of the individual call; they do not surface drift, provide time-zone-aware suggestions, support per-relationship intentionality, or offer reflective views of contact patterns. The implication for *keepclose* is positional: the project is designed as a *layer above* this infrastructure rather than as a replacement.

Launching external calling apps from within *keepclose* (FR4) directly expresses this layered relationship.

2.3.2 Wellbeing and meditation applications

A second category, visually and tonally relevant, is the consumer wellbeing application. Apps such as Calm *(Calm.com Inc., n.d.)*, Headspace *(Headspace Inc., n.d.)*, and Day One *(Bloom Built Inc., n.d.)* have established a design register for digital tools supporting psychological wellbeing: muted palettes, generous whitespace, gentle typography, and copy in a calm, low-pressure register. Interaction patterns are similarly restrained — no urgent notifications, no streaks foregrounded, no leaderboards.

The relevance is at the level of *register* rather than feature. These apps address internal mood regulation; *keepclose* addresses relational maintenance. But *keepclose*'s visual and tonal design — soft cream backgrounds, a single muted green for primary actions, gentle amber rather than red for the *Attention Needed* indicator — borrows from this category's vocabulary. The user research corroborates this: 84% of surveyed international students preferred a *gentle and supportive* or *neutral and practical* tone (Q24, §3.2.3).

2.3.3 Habit-tracking and gamified-engagement applications

The third category is the one *keepclose* is designed to reject rather than emulate. Habit-tracking and gamified apps — Streaks *(Crunchy Bagel Pty Ltd, n.d.)*, Habitica *(HabitRPG, Inc., n.d.)*, Duolingo *(Duolingo Inc., n.d.)*, and fitness-trackers in the same family — share a common pattern: progress measured numerically, streaks foregrounded, and a visual language emphasising performance and accumulation. These mechanics support behaviour change in their target domains *(Hamari, Koivisto and Sarsa, 2014)*, but apply a performance frame to the activity tracked.

For relational maintenance, this pattern is not merely unhelpful but counterproductive. The user research found that 44% cited *pressure to maintain streaks* as a deterrent, and 26% reported feeling guilty when contact slipped (§3.2.3). The streak mechanic — productive in language learning by adding extrinsic motivation to a behaviour users already want to perform — would function destructively in calling, converting intrinsic relational care into extrinsic obligation. The omission of streaks, scores, and gamified mechanisms in *keepclose* (NFR2) is the design's response.

2.3.4 Relationship-maintenance applications

The fourth and most directly comparable category is applications designed specifically for relationship maintenance: UpHabit *(UpHabit Inc., n.d.)* is representative. These recognise the same problem *keepclose* addresses — that everyday life erodes intentional contact with people who matter — and propose structurally similar solutions: per-contact frequency goals, reminders surfacing relationships that need attention, and contact logs over time.

Where they differ from *keepclose* is in framing. UpHabit is positioned as a *personal CRM* tool: contacts are treated as networking assets to be cultivated, frequency goals are framed in productivity terms.

The implication is twofold. The structural pattern is sound, and *keepclose* draws on it at the architectural level. But framing matters as much as structure: the international-student wellbeing context is poorly served by tools framed as networking optimisation, where the relationships in question are emotionally central rather than instrumentally useful. *keepclose* is positioned in the gap this framing difference creates.

2.4 Summary: identifying the gap

The literature (§2.2) and state-of-the-art (§2.3) together establish the gap *keepclose* is designed to occupy. International student loneliness and homesickness are documented experiences with real wellbeing implications (§2.2.1); voice and video calls with home networks are a validated coping practice and the dominant communication channel through which transnational connection is sustained (§2.2.2); and the broader HCI literature on awareness, presence, and relational technology offers design vocabulary for sustaining connection across distance (§2.2.3). The Peters (2023) framework provides the design lens through which these literatures translate into specific interaction decisions: autonomy as foundational, low-pressure framings over high-pressure ones, and sustained wellbeing over short-term engagement.

The state-of-the-art review supports the case for a calling-maintenance, wellbeing-supportive tool layered above existing communication infrastructure: none of the four categories reviewed occupies this space. General-purpose calling tools provide the communication action without supporting its maintenance over time. Wellbeing and meditation applications establish the right tonal register but address internal mood regulation rather than relational connection. Habit-tracking and gamified apps adopt patterns — streaks, scores, performance comparison — that are inappropriate for the relational domain and that the user research found to deter the population *keepclose* is designed to support (§3.2.3). Relationship-maintenance applications such as UpHabit share *keepclose*'s structural pattern but adopt a CRM-style framing poorly fitted to the international-student wellbeing context.

keepclose is positioned in the gap defined by these adjacent categories: a calling-maintenance layer above existing communication infrastructure, drawing tonally from the wellbeing-application register, rejecting the gamified mechanics of habit-trackers, and reframing the architectural pattern shared with relationship-maintenance applications away from networking optimisation toward wellbeing-supportive routine maintenance. The remainder of the dissertation describes how this design space was investigated through user research and translated into a prototype (Chapter 3), evaluated with international students (Chapter 4), and what the evidence implies for the design and future work (Chapter 5).

Chapter 3: What Was Done and How

This chapter describes the research, design and evaluation process by which the *keepclose* prototype was developed and assessed. It follows the actual sequence of project activities, so that each decision is presented alongside the evidence that justified it.

3.1 Approach overview

The project followed a human-centred design (HCD) process, in which the design responds to evidence about people and their practices rather than designer assumptions *(Sharp, Rogers and Preece, 2019)*. Within HCD, the design was anchored in Peters' (2023) wellbeing-supportive design perspective (Chapter 2), which emphasises autonomy-respecting, low-pressure interactions for systems intended to support psychological wellbeing over time. This perspective shaped both what was designed (a routine-maintenance layer rather than a behaviour-change system) and how it was evaluated (whether users found the prototype supportive, not only whether they could complete tasks).

The chapter's activities map onto the five objectives set out in the project proposal. Objective 1 (literature review) is covered in Chapter 2; its outputs feed the requirements derived in §3.3. Objective 2 — understanding the target users — was addressed through a 50-respondent online survey conducted via Google Forms (§3.2). Objective 3 — deriving the prototype's information architecture and interaction flows — was addressed by translating the survey findings and literature into functional and non-functional requirements (§3.3) and a prototype information architecture (§3.4). Objective 4 — producing the prototype — resulted in a high-fidelity Figma prototype of seven screens spanning three end-to-end user journeys (§3.5). Objective 5 — evaluation — was addressed through scenario-based walkthroughs with ten international students (§3.6). The proposal's planned post-evaluation iteration was subsequently scoped as optional; its implications are revisited in §3.6 and the conclusions chapter.

The output of the project is a design artefact rather than a deployed application — a deliberate scoping decision. The project investigates whether a particular kind of design — low-friction, calm, autonomy-supportive, routine-maintaining, layered on top of existing calling apps — could plausibly support international students in maintaining meaningful contact. That investigation is well served by a high-fidelity clickable prototype, evaluated with users in plausible scenarios; a fully implemented application would have committed substantial engineering effort to questions the prototype itself can already answer. Tooling reflected this scope: Google Forms for the survey, Figma for the prototype, and structured Word documents for the requirements, session notes and analysis.

Ethical considerations applied throughout, particularly in the user research and evaluation phases, and are described in §3.7.

3.2 User research

3.2.1 Aims

The user research aimed to provide an evidence base solid enough to inform the design decisions made in the subsequent prototyping phase. To be useful in this role, it had to

surface findings on three specific topics the literature alone could not answer at the granularity the design needed. First, it had to characterise *connection drift* in the lives of international students — how often it happens, what causes it, and what students do (and feel) when it does. Second, it had to test whether voice/video calling specifically — rather than messaging or other channels — is valuable enough to warrant a calling-specific design. Third, it had to surface students' preferences and concerns about the kinds of features such an app might offer, so the design could be shaped around what would be welcomed and against what would be rejected. These three lines of inquiry directly serve Objective 2 of the proposal.

3.2.2 Method

A 50-respondent online survey was the chosen instrument. Two factors supported this choice. First, the project's evaluation phase (§3.6) was already designed as a qualitative, scenario-based activity with international students, in which open-ended questions and observation would yield the depth that interviews would have provided at the research stage. Second, a survey is well suited to the kinds of questions the research needed to answer at this point — patterns and prevalence across a sample, comparative ratings of features, and self-reported attitudes — for which closed-form responses are appropriate *(Sharp, Rogers and Preece, 2019).*

The survey was implemented in Google Forms and open to international students at any institution; recruitment was via informal peer networks and student-facing online channels, on a voluntary basis. The instrument was anonymous: no identifying information (name, email, institution, programme code) was collected. The participant-facing wording avoided medical or clinical framing of homesickness and loneliness, in line with the wellbeing-sensitivity mitigation noted in the proposal.

The survey was organised into five sections:

1. **About you** — light demographics: age range, level of study, time spent abroad, and current living situation.
2. **How you currently keep in touch** — current calling and messaging practices, primary contacts at home and locally, platforms used, frequency, and whether calls are planned or spontaneous.
3. **When contact starts to drift** — the prevalence of drift, its main contributors, students' typical responses, the emotions associated with it, and the perceived effort of making calls.
4. **How important calling feels** — the perceived importance of voice/video calls, calls compared to messaging, and the perceived impact of drift on wellbeing.
5. **Possible app features** — comparative helpfulness ratings for seven candidate features, deterrents, attitudes to automatic call-history access, preferred tone, and an open free-text question.

The full questionnaire and the response distributions for each question (charts) are in Appendix A. Ethical considerations relating to the survey are described in §3.7, alongside those for the evaluation phase.

Headline quantitative findings from the survey are summarised in Figures 3.1–3.4 below; the implications drawn from these findings are discussed in the next section (§3.2.3).

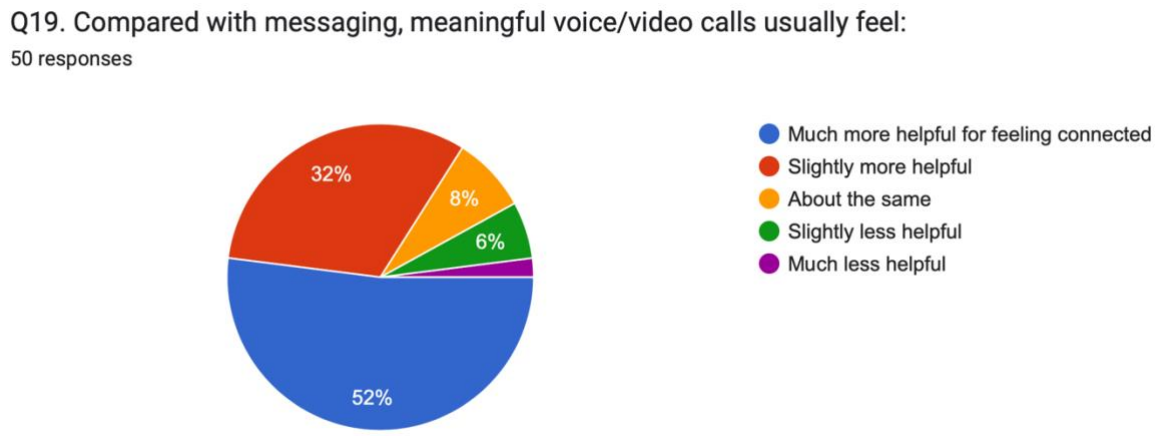

Figure 3.1: Respondents' rating of voice/video calls compared to messaging for feeling connected (n = 50).

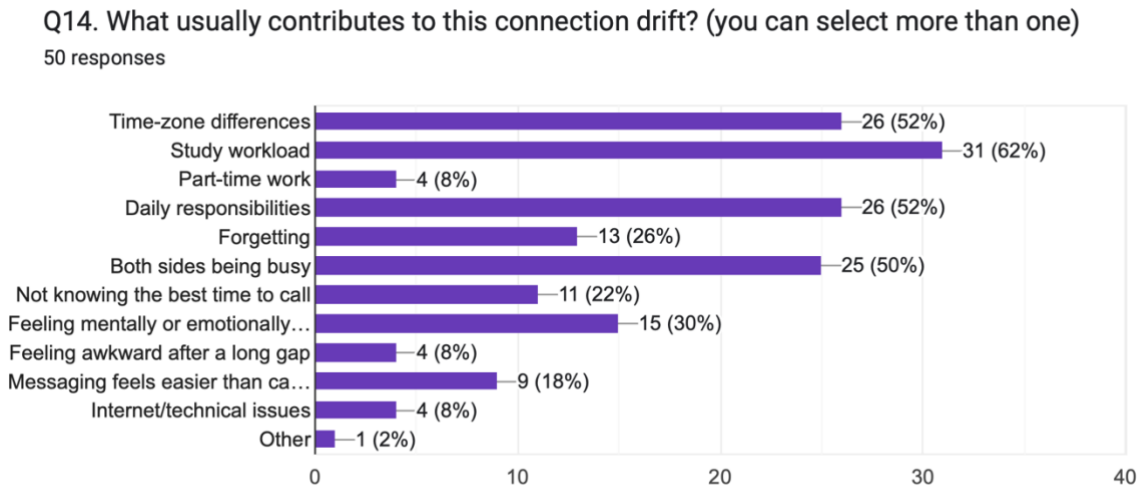

Figure 3.2: Self-reported contributors to connection drift across the 50-respondent sample.

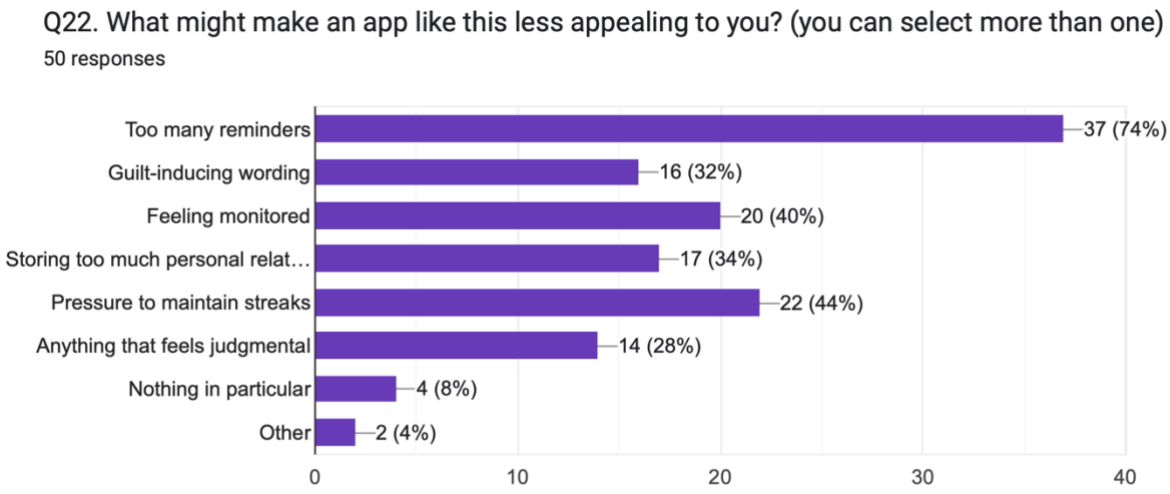

Figure 3.3: Self-reported features that would deter respondents from using an app designed to maintain regular meaningful calls with friends and family.

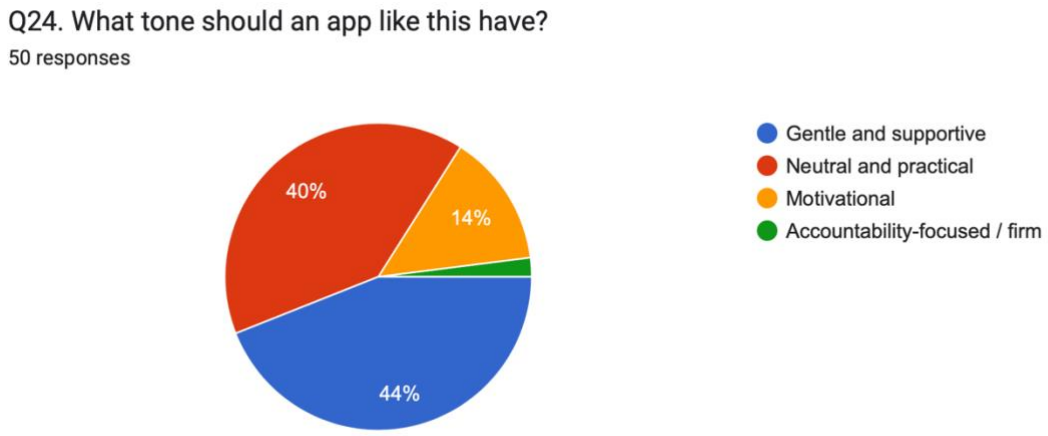

Figure 3.4: Respondents' preferred tone for an app designed to maintain regular meaningful calls with friends and family.

3.2.3 Findings used to drive the design

The full quantitative results, including the Google Forms charts for each survey question, are in Appendix A. The raw response data is available as Supplementary Material. This section reports only the findings that directly drove design decisions in §3.3 onwards; the implications are stated alongside each finding rather than left to the reader to infer.

The findings established that calling specifically is central to the population the project is designed for. A clear majority of respondents rated voice/video calls as important for feeling

connected to people who matter to them (84% rated 4 or 5 on a 5-point scale, Q18), and the same proportion rated calls more helpful than messaging for feeling connected (84%, Q19). This shaped the project's calling-specific framing — rather than a generic messaging or wellbeing-tool framing — at the requirements stage.

Connection drift emerged as widespread and emotionally consequential. 90% of respondents reported experiencing drift at least sometimes (Q13), and 76% agreed (≥3 on a 5-point scale) that drift negatively affects their sense of connection or wellbeing (Q20). The most common contributors (Q14) were study workload (62%), time-zone differences (52%), and daily responsibilities (52%); 50% also cited both sides being busy and 30% cited feeling mentally or emotionally unavailable. When drift occurs, the most common response was to switch to messaging (68%, Q15), suggesting calls are routinely replaced by lower-effort channels rather than rescheduled. Emotionally, drift was associated with feeling disconnected (46%), lonelier (42%), more homesick (38%), and guilty (26%) (Q16). These findings shaped the framing of the project as addressing a routine-maintenance problem rather than an episodic one, and located study workload and time-zone uncertainty as the two practical barriers the design needed to reduce.

Feature preferences (Q21) shaped the design's specific direction. All seven candidate features rated above 3.0 (mean score, on a 5-point helpfulness scale), with gentle reminders (mean = 3.90), time-zone suggestions (mean = 3.88), and quick launch into existing calling apps (mean = 3.86) the three highest-rated. These three became the core of the prototype's interaction model.

Deterrents (Q22) were equally consequential. The most-cited was *too many reminders* (74%), followed by *pressure to maintain streaks* (44%), *feeling monitored* (40%), *storing too much personal relationship data* (34%), *guilt-inducing wording* (32%), and *anything that feels judgmental* (28%). The asymmetry between the most-rated feature (gentle reminders, 74% helpful) and the most-cited deterrent (too many reminders, 74%) shaped the project's thinking about reminder design: reminders are simultaneously the most welcome and the most dangerous design element. Attitudes to privacy reinforced the same direction: only 26% would want the app to access call history automatically, with 38% answering *No* and 36% *Maybe* (Q23). On tone (Q24), 84% preferred *gentle and supportive* or *neutral and practical*, and only 14% preferred *motivational* — pointing the design toward a register distinct from conventional habit-trackers.

These findings are revisited in §3.3, mapped onto the requirements they shaped.

3.2.4 Limitations of the user research

The survey approach has limitations. The sample was self-selected: respondents who completed the survey may be more reflective about their connection patterns than international students in general, which could skew emotional and behavioural responses. The data is self-reported and therefore subject to recall and social-desirability effects, particularly on questions about how often calls happen and how drift makes respondents feel. Recruitment relied on informal peer and student online channels, constraining how broadly the sample reflects the international-student population. Although the sample includes variation in home country, course level, and length of time abroad (Appendix A), this variation was not systematically sought, and the sample is therefore indicative rather than representative. Finally, although n = 50 is a reasonable size for an undergraduate project, it

does not support fine-grained subgroup analysis — for example, comparing newly arrived students with those abroad for several years. These limitations are revisited in §3.6.7 alongside the evaluation, and in the conclusions chapter.

3.3 Design requirements

3.3.1 Synthesis approach

Translating the user research findings into a concrete brief for the prototype was carried out as an explicit synthesis step between the user research and prototyping phases. This served two purposes. First, it forced explicit synthesis between evidence and design: each requirement is traceable to one or more specific survey questions. Second, it provided a stable reference against which later design decisions could be checked, preventing drift between what the research shaped and what the prototype actually contained.

The requirements are split into functional requirements (FRs), describing what the system must do, and non-functional requirements (NFRs), describing how it must behave or feel *(Sharp, Rogers and Preece, 2019)*. For a wellbeing-maintenance tool, NFRs carry unusual weight: failures in tone, default behaviour, or perceived surveillance would not merely degrade usability — they would actively undermine the project's purpose. They are therefore treated as first-class constraints, not as polish applied after the functional design is set.

Each requirement is presented with a description, the user-research source(s) supporting it, a one-line rationale, and a priority (High / Medium / Low). Priorities reflect how strongly the evidence supports the requirement, how central it is to the core interaction model, and the consequences of leaving it out of the prototype. High-priority items are those without which the prototype would not be a meaningful test of the project's design hypothesis; Medium-priority items support the experience but could be partially scoped; Low-priority items are useful additions whose absence would not invalidate the prototype.

3.3.2 Functional requirements

The seven functional requirements describe the behaviours the prototype must support to test the project's core hypothesis: that a low-effort, autonomy-respecting calling layer can help international students maintain meaningful contact in the face of connection drift. Together, they cover noticing drift (FR3), deciding to call (FR1, FR2), making a call (FR4), recording it (FR5), and reflecting on patterns (FR6), with FR7 ensuring the structural model can adapt to individual identity.

ID	Requirement	Source	Rationale	Priority
FR1	Users can add important contacts and set a desired call frequency per contact (e.g. once a week, a few times a month).	Q21-Feature 5 (3.66/5, 56% helpful); Q12	Calls are mostly spontaneous (50%) or loosely routine (38%); frequency-setting provides a personal, low-pressure structure where one is otherwise absent.	High

ID	Requirement	Source	Rationale	Priority
FR2	The app displays suggested time windows to call a contact, taking into account their approximate time zone.	Q21-Feature 2 (3.88/5, 66% helpful); Q14 (52% time-zone differences)	Time-zone uncertainty was the second most-cited drift cause; suggestions reduce the friction of knowing when to call.	High
FR3	The app delivers gentle, infrequent reminders when a contact's call frequency falls below the user's stated goal.	Q21-Feature 1 (3.90/5, 74% helpful); Q22 (74% over-reminding)	The highest-rated feature, but also the most-cited deterrent if misimplemented; reminder frequency and tone must be carefully controlled (see NFR1, NFR4).	High
FR4	A single tap on a contact card launches the user's preferred calling app (e.g. WhatsApp, FaceTime, Telegram) directly.	Q21-Feature 3 (3.86/5, 68% helpful); Q15 (68% switch to texting)	The app must reduce friction, not add to it; direct launch removes the barrier of switching apps and reinforces quick action.	High
FR5	After a scheduled or attempted call, users can optionally log the outcome: Called, Missed, or Postponed.	Q21-Feature 7 (3.54/5, 58% helpful); Q22 (34% data-storage concern)	Useful for tracking patterns but must be kept lightweight; making logging optional respects users who find it burdensome.	Low
FR6	Users can view a simple visual summary of contact frequency per person over a rolling period (e.g. past 4 weeks), with no scores, streaks, or comparisons.	Q21-Feature 4 (3.78/5, 62% helpful); Q22 (44% opposed to streaks)	Users want to reflect on patterns but reject gamified representations; neutral visualisation is appropriate.	Medium
FR7	Contacts can be organised into user-defined Circles; default Circles of 'Home' and 'Local' is	Q21-Feature 6 (3.56/5, 62% helpful); Q25 (open-text identity concern)	A fixed 'Home' label may feel alienating to students who have rebuilt their sense of home; editable labels cost nothing to implement and prevent a real harm.	Medium

ID	Requirement	Source	Rationale	Priority
	offered but the labels are editable.			

Table 3.1 — Functional requirements

3.3.3 Non-functional requirements

The non-functional requirements were derived primarily from the deterrent and tone questions (Q22, Q23, Q24) and the qualitative critique in Q25. They define the behavioural envelope within which the FRs must operate. Several encode design prohibitions — things the app must *not* do — because the survey evidence on what students would reject was as strong as the evidence on what they would value.

ID	Requirement	Source	Rationale	Priority
NFR1	Reminder notifications must be configurable in frequency by the user, with a conservative default (e.g. at most once per contact per week).	Q22 (74% 'too many reminders')	The single most-cited deterrent; a design that sends too many reminders will be uninstalled regardless of other qualities.	Medium
NFR2	The app must not use streak mechanics, scores, league tables, or achievement badges. No element should imply a competitive or achievement-based framing.	Q22 (44% 'pressure to maintain streaks'; 28% 'anything judgmental'); Q16 (26% feel guilty when drift occurs)	Habit-tracker and fitness-app metaphors are inappropriate for an emotionally sensitive domain and would compound rather than reduce guilt.	High
NFR3	The app must not automatically access or read device call logs. Any contact or call data must be manually entered by the user.	Q23 (38% No, 36% Maybe on automatic call history)	A near-majority are resistant; automatic access would deter a significant proportion of potential users and creates an avoidable privacy risk.	High
NFR4	All written copy — reminders, status labels, onboarding text — must use	Q22 (32% 'guilt-inducing wording'); Q24 (84%	Language is a primary interaction surface for this app; tone design is as important as visual design.	High

ID	Requirement	Source	Rationale	Priority
	gentle, neutral, or encouraging language. Wording that implies failure, inadequacy, or judgment is prohibited.	gentle/neutral tone)		
NFR5	The minimum viable interaction path (seeing a contact's status and launching a call) must require no more than 2 taps from the home screen.	Q15 (68% switch to texting); Q14 ('not knowing the best time to call')	Effort is a primary reason calls are avoided; the app must compete with the near-zero effort of opening WhatsApp directly.	High
NFR6	Contact data (names, frequency settings, call logs) must be conceptually local to the device by default and must not be transmitted externally without explicit user consent.	Q22 (40% 'feeling monitored'; 34% 'storing too much personal data')	Relationship data is sensitive; a Figma prototype does not transmit data, but the principle must be stated and designed into the interaction flows.	Medium
NFR7	Feature depth must be opt-in. The default view should be minimal; tracking, statistics, and frequency goals should be available but not foregrounded.	Q25 (open-text suggestion); Q22 (general surveillance concern)	Users who find tracking unhelpful should be able to use the app without engaging it; opt-in depth respects autonomy.	Medium

Table 3.2 — Non-functional requirements

The most consequential pattern across the two tables is the asymmetry between FR3 (gentle reminders, the highest-rated feature) and NFR1 (configurable reminders, derived from the highest-cited deterrent). The same survey item that motivates the prototype's most popular feature is also its single largest design risk. This asymmetry is what makes wellbeing-maintenance design genuinely different from productivity or habit-tracking design: a feature that is excellent at one frequency becomes intolerable at another. The joint constraint imposed by FR3 and NFR1 is what makes the prototype's reminder behaviour conservative by default rather than aggressive.

The information architecture and core user journeys derived from these requirements are described in §3.4.

3.4 Information architecture and core user journeys

3.4.1 Information architecture

The requirements derived in §3.3 were translated into a five-area information architecture (IA), summarised in Table 3.3. The five functional areas are not the same as the seven physical screens of the prototype: some areas are realised by a single screen, others by a screen with multiple states (such as the Add Contact form, which has both an empty and a filled state), and one area — Settings — was deliberately scoped down to its Add Contact subset, for reasons explained at the end of this sub-section.

#	Functional area	Purpose	Primary requirements served
1	Contacts	Central overview of important contacts and their connection status at a glance	FR3, FR7, NFR2, NFR5
2	Contact Detail	Individual contact profile: frequency goal, last called, time-zone helper, call history, call launch button	FR1, FR2, FR4, FR5
3	Quick Call Launch	One-tap launch into the user's preferred calling app (e.g. WhatsApp, FaceTime)	FR4, NFR5
4	Connection History	Calm, non-comparative view of recent call frequency across contacts	FR6, NFR2
5	Add Contact / Settings	Add and edit contacts, set frequency goals, configure Circles	FR1, FR7

Table 3.3 — Five functional areas of the keepclose IA

Top-level navigation is anchored by a three-destination bottom navigation bar (Contacts, History, Settings), present on the principal screens. Quick Call Launch (area 3) is intentionally not a destination but a modal sheet summoned from Contact Detail — its design is to feel like a system-level action rather than a destination users navigate to. Add Contact is reached via a floating action button on Contacts rather than via the Settings tab, because adding a contact is the most common configuration action and burying it inside Settings would have violated NFR5 (the two-tap principle).

The principal screens are illustrated in Figures 3.5–3.11 (presented in §3.5.2 alongside their design rationale) and were built in Figma at high-fidelity. Figure 3.5 shows Contacts; Figure 3.6 shows Contact Detail; Figure 3.7 shows the Quick Call Launch sheet overlaid on Contact Detail; Figure 3.8 shows the History screen; Figures 3.9 and 3.10 show the empty and filled states of the Add Contact form; and Figure 3.11 shows Contacts after a new contact (Sara)

has been added. The seven screens realise the five functional areas through three single-state screens, one modal sheet, one two-state form, and one home screen captured in two states.

The broader Settings area (reminder preferences, per-contact frequency editing post-creation, Circle-label editing, data transparency notes) was not implemented as a distinct screen. This was a deliberate scoping decision: none of the three core user journeys (§3.4.2) — and therefore none of the three evaluation scenarios (§3.6) — exercises post-setup configuration. Building the settings screens would have absorbed prototyping time without producing evaluable interactions, while the principles behind those settings (opt-in defaults, editable Circle labels, conservative reminder frequency) are tested through the journeys that do exist. The trade-off is revisited as future work in the conclusions chapter.

3.4.2 Three core user journeys

Three end-to-end journeys were prototyped as fully wired, walkable flows in Figma. Together they cover the maintenance loop the project is built around — notice that contact has slipped, decide and act on it, and reflect on the resulting pattern — plus the bootstrap step of adding a person to maintain contact with. The three journeys map one-to-one onto the evaluation scenarios in §3.6, so the prototype supports the evaluation as a coherent whole rather than as a set of disconnected screens.

Journey 1 — Drift to Call (Figures 3.5 → 3.6 → 3.7). The user opens the app and sees a contact card showing it has been some time since they last called that person. They tap the card to reach Contact Detail, where the time-zone helper indicates that it is currently a good time to call. They tap *Call now*, which presents the Quick Call Launch sheet, and select their preferred app. This journey exercises FR2, FR3, FR4, NFR4, and NFR5, and is the prototype's primary use case.

Journey 2 — *Add a Contact* (Figures 3.5 → 3.9 → 3.10 → 3.11). The user taps the floating action button on Contacts, completes the Add Contact form (name, frequency goal, Circle, preferred calling apps), and saves. They are returned to Contacts with the new contact visible. This journey exercises FR1, FR7, and NFR5, and establishes the routine-maintenance framing by making the user's calling intention explicit at setup.

Journey 3 — *Review Patterns* (Figures 3.5 → 3.8). The user taps *History* in the bottom navigation to reach Connection History, reviewing the calm, non-comparative summary of recent contact patterns. This journey is short by design — the screen is intended to be glanced at rather than worked through — and exercises FR6 and NFR2. Because users may relate to this screen emotionally rather than instrumentally, the evaluation (§3.6) was designed to ask not only whether participants could use it but how it made them feel.

The three journeys collectively exercise all seven functional requirements at least once, and the five high-priority non-functional requirements in the form of design decisions made within the journeys. NFR6 and NFR7, which describe principles rather than journey-level interactions, are addressed at the design-decision level in §3.5, where the detailed design choices that shape each screen are also described.

3.5 Prototype design and rationale

3.5.1 Tool and fidelity choice

The prototype was built in Figma and presented as a high-fidelity, fully wired clickable design rather than a low-fidelity sketch. The choice was driven by the nature of the design hypothesis itself: the project is testing whether a low-friction, calm, autonomy-respecting visual and interaction language can support international students in maintaining meaningful contact, and that hypothesis is difficult to evaluate at low fidelity, because tone and visual restraint are largely invisible in greyscale wireframes.

The trade-off is acknowledged. The HCI literature recommends low-fidelity sketching for generating and discarding ideas cheaply *(Sharp, Rogers and Preece, 2019)*, and skipping that stage forecloses the rapid divergent exploration paper prototypes support. To partially compensate, alternative interaction patterns (e.g. dashboard-first vs contact-first home, mandatory vs optional logging, achievement-style vs neutral history visualisation) were considered and discarded in writing before any Figma frame was created. The prototype was then iterated continuously during construction in response to internal review, although no separate post-evaluation iteration was undertaken (§3.6).

3.5.2 Screen-by-screen rationale

The following sub-sections describe the five functional areas in the order a typical user would encounter them, covering each screen's purpose, principal design decisions, and the requirements served.

Contacts (Figures 3.5 and 3.11). The app opens directly to the contacts list. This is intentionally a contacts home, not a dashboard: the survey indicated that the people users wanted to maintain contact with were highly concentrated (parents and siblings being the most frequently named primary contacts), making the appropriate cognitive frame *people, not metrics*. The screen is organised by three top-level tabs (*Home* / *Local* / *All*) corresponding to the user's two default Circles plus an aggregate view (FR7, NFR5). Within the active tab, contacts are grouped under two soft section headings — *Attention Needed* and *On Track* — corresponding to whether the last call falls within the user's stated frequency goal. Cards in the *Attention Needed* group use a muted amber accent and small status dot rather than a red badge or warning icon (NFR4). Each card displays a brief time-zone status (e.g. *Good · 2pm there*, *Late · 3am there*), implementing FR2 at list level so the user can decide whether to call without entering the detail screen. The floating action button for adding a new contact (FR1) is positioned bottom-right, and the bottom navigation bar exposes the three top-level destinations described in §3.4.1.

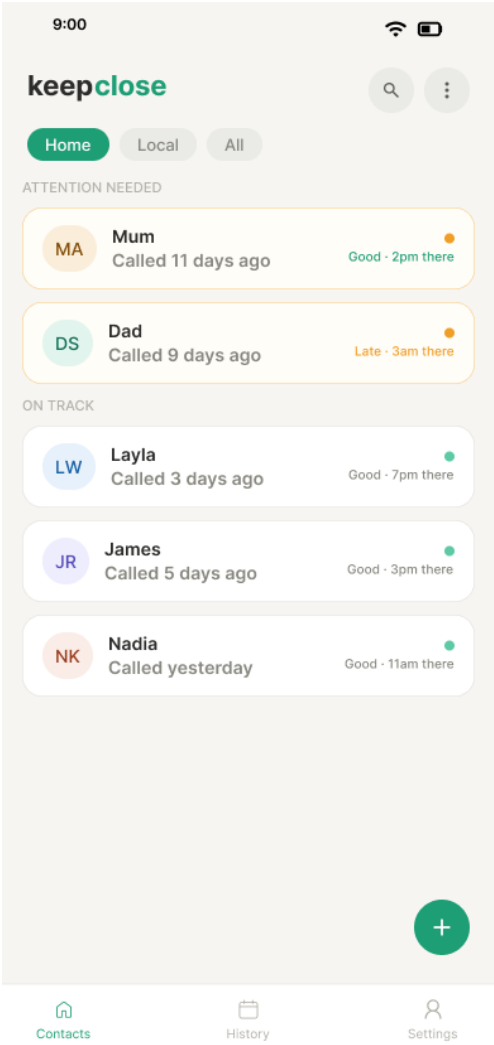

Figure 3.5: Contacts screen, showing the Home Circle with two contacts under Attention Needed and three under On Track.

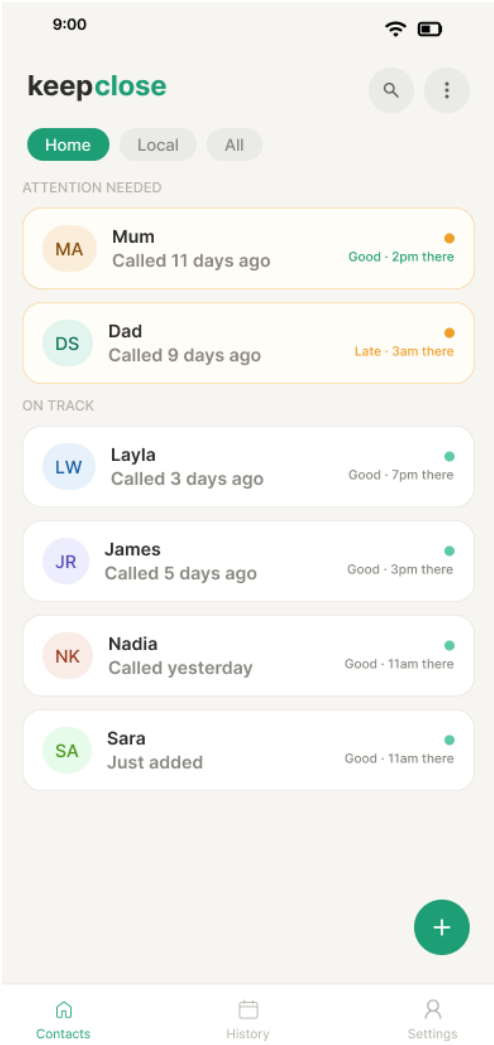

Figure 3.11: Contacts screen after the new contact (Sara) has been added.

Contact Detail (Figure 3.6). This is the screen on which the app's central decision — whether and when to call — is taken. Its layout is deliberately uncluttered. The contact's name, avatar, and Circle tag sit at the top, followed by the time-zone helper card showing the contact's local time, the time difference, and a one-word status (FR2). The primary action — a large green *Call now* button — is by far the most visually prominent element on the screen, satisfying FR4 and NFR5: from app launch, the user reaches a calling app in two taps (Contacts → Contact Detail → *Call now*). Beneath the primary action, *Log a call instead* is rendered as a smaller secondary affordance (FR5), making logging available without making it feel obligatory. The user's stated calling goal is shown with an inline edit affordance (FR1), and the four-week call history is displayed as four small tiles (FR6) — ticked or empty, with no scoring or comparison — accompanied by a single neutral sentence (e.g. *2 calls in the last month*) in plain language (NFR2).

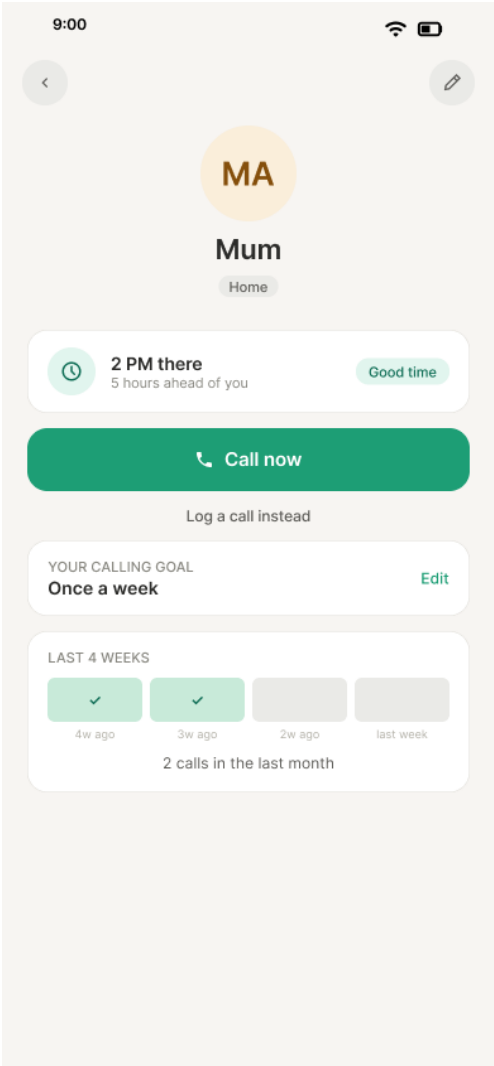

Figure 3.6: Contact Detail screen for Mum, showing the time-zone helper, the primary Call now action, the user's stated calling goal, and a four-week summary of recent calls.

Quick Call Launch (Figure 3.7). When the user taps *Call now*, a modal sheet appears showing the calling apps the user has associated with that contact, and a single tap launches the chosen app (FR4). The sheet slides up from the bottom of the screen rather than taking the user to a new page, signalling that the action is launching out to another application — reinforcing that *keepclose* is a layer over the user's existing calling tools rather than a replacement for them. This responds directly to the survey finding that students already use a small set of established calling platforms; adding another would have created friction rather than removing it.

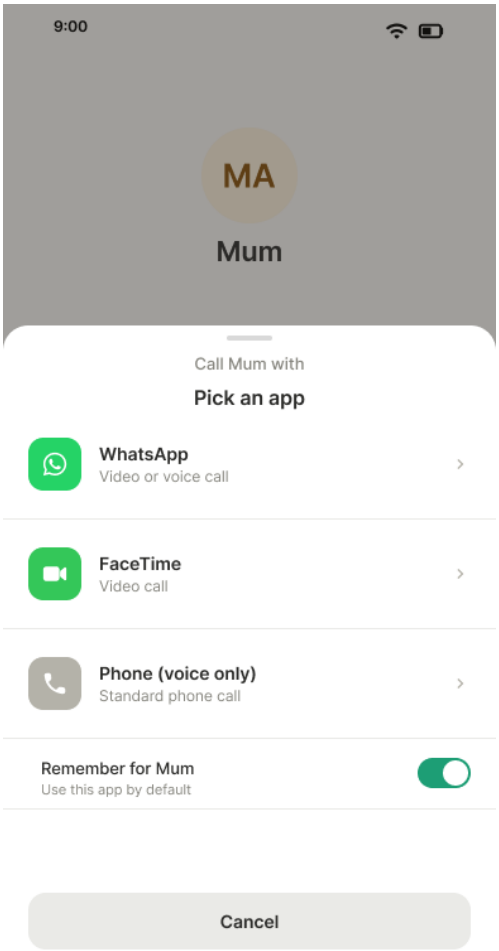

Figure 3.7: Quick Call Launch sheet, presented when the user taps Call now.

History (Figure 3.8). A secondary screen, accessible from the bottom navigation, providing a calm view of recent contact patterns. The title and subtitle — *History* and *A quiet look at contact patterns* — frame the screen as reflective rather than evaluative. Each contact appears as a row containing a four-tile pattern (one tile per week) and a neutral count (e.g. *2 calls*); there are no streaks, scores, rankings, or aggregate connection-health metrics (FR6, NFR2). A range toggle allows the user to switch between four-week and eight-week views.

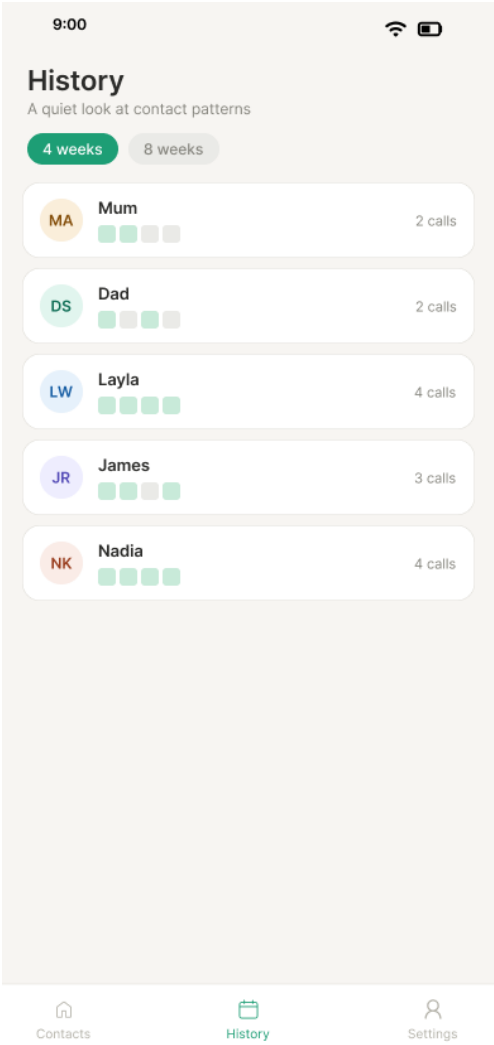

Figure 3.8: History screen, showing a calm, non-comparative summary of recent contact patterns across contacts.

Add Contact (Figures 3.9 and 3.10). Reached via the floating action button on Contacts, this is the only configuration screen prototyped. The form requires only what is necessary for the three core journeys: name, phone number, the calling apps the user wants associated with this contact (toggleable chips), the Circle (HOME / LOCAL), the contact's time zone, and the calling goal. The time-zone field includes the helper sentence *So we can suggest good times to call*, attaching the user's input to the user-visible benefit (FR2). The calling goal offers three presets (*Once a week*, *Every two weeks*, *Once a month*) plus a *Custom* option, balancing low-effort defaults against autonomy (FR1).

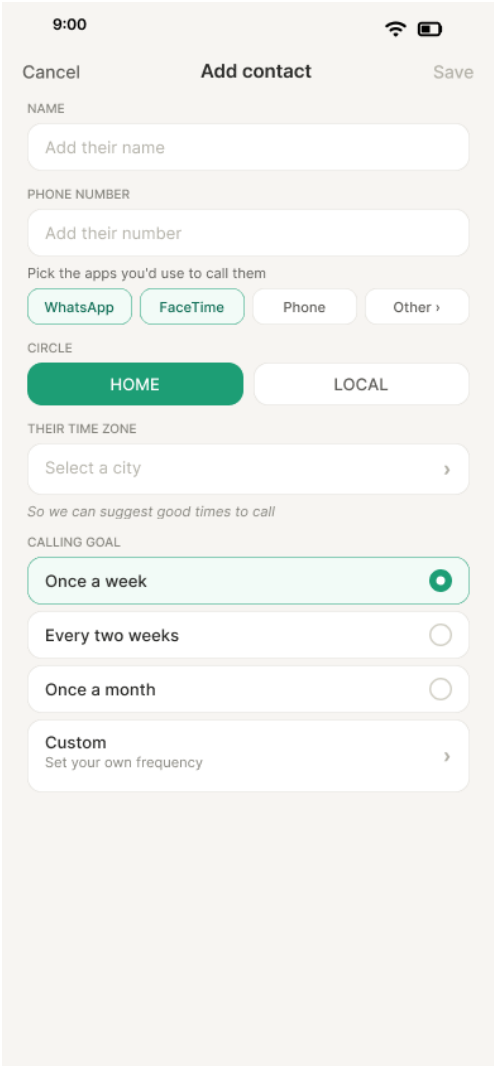

Figure 3.9: Add Contact form in its empty state, with defaults for the calling apps (WhatsApp, FaceTime), Circle (Home), and calling goal (Once a week).

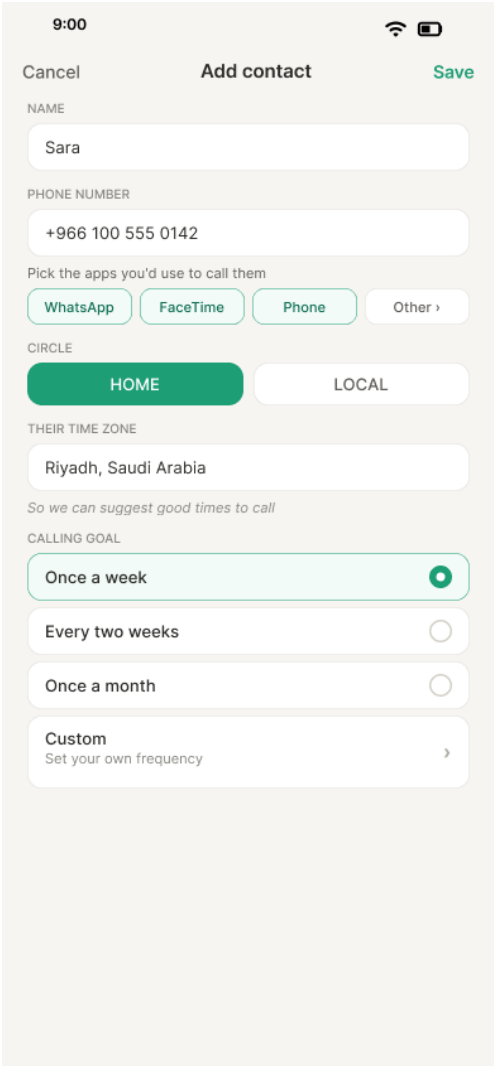

Figure3.10: Add Contact form in its filled state.

3.5.3 Cross-cutting design decisions

Three design decisions cut across multiple screens.

Visual and typographic tone. The prototype uses a restrained palette — soft cream backgrounds, a single muted green for primary actions and confirmation, gentle amber for *Attention Needed* highlights, and neutral greys for secondary text — together with generous whitespace and a single sans-serif typeface at conservative weights. The aim was to make the app *feel* unlike a productivity or fitness tool. The visual language is closer to the calm-app category (journaling and meditation apps) than to the dashboard category (step counters and productivity trackers), and this choice is itself part of the design hypothesis being evaluated.

Microcopy and the *Attention Needed* tension. Written copy across the prototype is short, low-pressure, and avoids both clinical and cheerful registers (NFR4). One specific microcopy

choice — the *Attention Needed* section heading on Contacts — sits at the boundary of the design's tone. Surfacing drift is one of the prototype's central functions, and some phrasing is needed to indicate which contacts the system thinks the user might wish to call; a stronger phrase (e.g. *Overdue*) would have been inappropriate, but a softer phrase (e.g. *You haven't spoken in a while*) would have been long for a section heading. *Attention Needed* was selected as the most concise framing that did not imply failure or judgment, with the understanding that the evaluation would test whether participants found the wording acceptable. Findings on this point are reported in Chapter 4.

Deliberate omission of streaks, scores, and gamification. The most consequential design decision is one of restraint: the prototype contains no streaks, scores, badges, aggregate connection-health metrics, comparisons between contacts, or implicit rankings. This is grounded in the survey evidence (44% identified *pressure to maintain streaks* as a deterrent and 26% reported feeling guilty when drift occurs) and in Peters' (2023) wellbeing-supportive design framework, which identifies autonomy and the avoidance of pressure as preconditions for technology that supports rather than undermines psychological wellbeing. The point was reinforced by one survey respondent (R15), whose Q25 free-text answer noted that *"some of the things can only add to the international student loneliness, like planning calls, setting streaks"* and that *"putting contacts into 'Home' and 'Local' can be upsetting, because if the student plans to stay in the country where they studied, this country becomes their home too."* This single response is the principal qualitative source for NFR2 (no streaks), FR7 (editable Circle labels), and NFR7 (opt-in feature depth) — an example of how a single richly considered respondent answer can do disproportionate evidential work in a primarily quantitative survey.

3.6 Evaluation methodology

3.6.1 Aims

The evaluation had two intertwined aims, addressing Objective 5 of the proposal. The first was the standard usability question: could international students complete the prototype's three core journeys with low cognitive effort, and did the interaction model behave as intended? The second was the project-specific question, central to a wellbeing-focused project: did the prototype *feel* the way the design intended — calm rather than judgmental, supportive rather than surveilling, autonomy-respecting rather than obligation-creating? The second matters because a prototype that is technically usable but emotionally wrong would not merely fail to support the project's wellbeing aim — it would actively undermine it. The evaluation method therefore had to surface both task-level performance and emotional response in the same session.

3.6.2 Method: scenario-based walkthroughs with structured feedback

The chosen method was scenario-based walkthroughs with structured open-ended feedback — a qualitative usability method in which participants are presented with realistic situations rooted in the target context and asked to act on them using the prototype, while the researcher observes task performance and follows up with open-ended questions *(Sharp, Rogers and Preece, 2019)*. Two alternative methods were considered and rejected. Heuristic evaluation was discounted because it would have produced expert-derived findings rather than evidence about how the user group interprets the prototype, and because heuristics such as Nielsen's are calibrated for productivity tools rather than the autonomy and tone concerns at the centre

of this project. A pure usability test using metrics such as time-on-task and the System Usability Scale was discounted for related reasons: such metrics would have treated the prototype as a productivity tool whose merit depends on speed and learnability rather than emotional fit. Scenario-based walkthroughs produce qualitative data speaking to both aims at once, and the scenarios themselves can be derived from the user-research findings (Q14 and Q15), giving the evaluation a strong evidential grounding in the situations the prototype is intended to support.

3.6.3 Participants

Ten international students participated (P01–P10), recruited via informal peer networks and snowball sampling on a voluntary basis. The sample was diverse on the dimensions most relevant to the project: time spent abroad ranged from 3 months (P10) to 4 years (P06); home / host-country time-zone difference from 1 hour (P03) to 9 hours (P07); the cohort included undergraduate, postgraduate taught, PhD, and foundation-level students; and both iOS and Android users were represented (six and four respectively). This diversity mattered because several aspects of the design — particularly the time-zone helper and the Home / Local Circle labels — were likely to be experienced very differently by students at the extremes of these dimensions.

3.6.4 Procedure

Sessions ran for 25 to 34 minutes (median 30) and were conducted on 24 and 25 April 2026, in person where possible (eight sessions) and via video call where required (two sessions). Each session followed an identical structured procedure. After welcoming the participant, the researcher confirmed they had read the information sheet and returned the consent form, and verbally re-iterated the right to skip questions or withdraw at any time. Light demographic information (time abroad, time-zone difference, course level, phone operating system) was collected; no identifying information was recorded.

Each participant then walked through three scenarios in fixed order, corresponding to the three core user journeys (§3.4.2). Each scenario was read aloud verbatim from the scenario script (Appendix B). The scenarios placed the participant in plausible international-student situations rather than abstract usability tasks: in Scenario 1, a parent at home had not been called in some time; in Scenario 2, a sister had recently moved to another country; in Scenario 3, the participant was reviewing their recent contact patterns at month-end. For each scenario, the researcher observed task completion against a structured checklist of expected sub-steps (recording each as completed, partial-with-hint, or failed), noted spontaneous comments, and asked two or three open-ended post-task questions surfacing the participant's interpretation of the design — including questions about the *Attention Needed* framing on Contacts, the *calling goal* construct on Add Contact, and the emotional tone of the History screen.

After all three scenarios, two overall questions invited the participant to reflect on whether such an app could plausibly help international students stay connected, and what they would add or remove. The session ended with the researcher's reflection notes (top takeaways, surprises, and short tag keywords), filled in immediately after the participant left. The full session-note template is in Appendix C. Anonymised session notes for all ten participants are available as Supplementary Material.

3.6.5 Materials

The evaluation materials comprised the participant information sheet (Appendix D), consent form (Appendix E), scenario script (Appendix B), and session-note template (Appendix C). The prototype was presented as an interactive Figma prototype for in-person sessions and in shared-screen mode for video-call sessions, with the researcher resetting the prototype to Screen 1 between scenarios.

3.6.6 Analysis approach

The analysis was qualitative and structured rather than statistical. For each scenario, the task-completion data from the ten sessions was tallied to identify any sub-step that produced systematic difficulty, and participants' open-ended responses were grouped to surface common themes. The session-by-session tag keywords (e.g. *attention wording*, *time-zone helpful*, *home/local concern*, *wants priority per contact*) provided an initial coding layer, refined through close re-reading of the full notes for each participant. The approach is closer to a light thematic analysis *(Braun and Clarke, 2006)* than to formal coding: with ten participants and a structured note template, purpose-built qualitative analysis software (e.g. NVivo) would have added more setup overhead than analytical value, while a structured re-reading was sufficient to surface the patterns reported in the next chapter. Triangulation across the three data streams — task-completion observations, post-task responses, and researcher reflection — was used as the primary check on individual themes.

3.6.7 Limitations

The evaluation has limitations. The sample is not very large (n = 10) and self-selected, which limits generalisability — particularly to international students who would not volunteer for a research session about staying in touch with family. The scenarios are necessarily artificial: a participant imagining a call to their own mother is not in the same emotional state as someone actually placing the call, and emotional responses in the session may understate or misrepresent how the prototype would be experienced in real use. The evaluation is single-session and non-longitudinal; behaviours such as reminder fatigue or the long-term effect of viewing the History screen weekly cannot be assessed in a 30-minute walkthrough.

The proposal's planned post-evaluation iteration cycle was not undertaken; iteration was subsequently scoped as optional. Given time constraints and the scope of the analysis the single round produced, time was directed into the evaluation analysis rather than a partially completed second iteration. The findings reported in Chapter 4 describe the prototype as evaluated; the design improvements the evaluation implies are described in the conclusions chapter as future work.

3.7 Ethical considerations

Both phases of empirical work — the user research survey (§3.2) and the prototype evaluation (§3.6) — involved human participants and were therefore designed and conducted in accordance with Newcastle University's research ethics framework. The CSC3094 module operates under a standing ethical approval that covers low-risk undergraduate projects, meaning individual projects do not require separate ethics applications provided they operate within the conditions of that approval: adult voluntary participants, anonymous data, no

clinical or vulnerable-group framing, and no collection of sensitive personal information. The design of both empirical phases was kept within these conditions throughout.

Beyond this baseline, four specific ethical concerns shaped how the empirical work was carried out: voluntary and informed consent, anonymity and data handling, sensitivity of the topic, and the avoidance of unnecessary data collection.

Voluntary and informed consent. Both phases were conducted on a voluntary basis with no incentive offered. Survey respondents were presented with a short introduction explaining the study's purpose, what participation involved, the voluntary nature of participation, and how their responses would be used. Before any substantive questions, respondents were required to tick three confirmation items: that they were international students, aged 18 or above, and agreed to take part in the anonymous survey. Only respondents who confirmed all three proceeded to the main survey. Evaluation participants received the participant information sheet (Appendix D) before the session and returned the signed consent form (Appendix E) at the start of the session. The information sheet and consent form covered the purpose of the research, the format and length of the session, the right to withdraw at any time without giving a reason, the nature of the data collected and how it would be stored, and the possibility that anonymous quotes might appear in the dissertation. The right to withdraw and to skip questions was reiterated verbally at the start of each session.

Anonymity and data handling. No identifying information (name, email address, programme code, or institution) was collected in either phase. Survey responses were stored as anonymous Google Forms submissions. Evaluation participants were assigned codes (P01–P10), and all session notes, post-task answers, and researcher reflections were recorded under those codes only. Demographic information collected at the start of each session was limited to four light items (time abroad, time-zone difference, course level, phone operating system) directly relevant to interpreting the session. Research files (survey export, session notes, analysis documents) were stored on the researcher's password-protected personal device.

Sensitivity of the topic. Homesickness and loneliness can be personally meaningful and, for some students, distressing. Both phases were therefore designed to discuss these topics in the abstract — as design problems faced by international students as a population — rather than to elicit personal disclosures. Survey questions focused on patterns of behaviour and attitudes toward features, not on respondents' own emotional histories. The evaluation scenarios involved hypothetical situations (calling "your mum", adding "your sister Sara") rather than asking participants to recount their own relationships, and participants were free to engage at whatever level of personal connection they chose. The research deliberately avoided clinical or medical framing of homesickness and loneliness, in line with the wellbeing-sensitivity mitigation noted in the project proposal.

Avoidance of unnecessary data collection. A deliberate choice was made not to audio- or video-record the evaluation sessions. The information sheet stated that no recording would take place unless the participant explicitly consented; in practice no participant was asked for recording consent and no sessions were recorded. The structured session-note template (§3.6.4) was sufficient to capture the data the analysis required, and the absence of recordings reduced both the sensitivity of the data held and the cognitive load on participants. This is a methodological cost — reliance on contemporaneous notes loses some nuance — but the trade-off was judged appropriate for the topic and scale of the study.

The combined effect is that the project's empirical work generated only the data necessary for the research questions, in a form that could not be linked back to any individual participant. The researcher was the sole handler of the data throughout. No ethical incidents arose during either phase.

3.8 Summary

This chapter has described the human-centred design process by which the *keepclose* prototype was developed and evaluated. A 50-respondent survey of international students provided the evidence base (§3.2), translated into seven functional and seven non-functional requirements (§3.3), an information architecture organised around five functional areas and three core user journeys (§3.4), and a high-fidelity Figma prototype of seven screens whose design decisions were grounded in the survey findings and in Peters' (2023) wellbeing-supportive design framework (§3.5). The prototype was evaluated through scenario-based walkthroughs with ten international students using a structured procedure designed to surface both task-level usability and how the prototype felt to use (§3.6). Ethical considerations — voluntary and informed consent, anonymity, sensitivity to the topic, and minimisation of data collection — applied across both empirical phases (§3.7).

The findings produced by the evaluation — task-completion patterns, the participants' interpretations of the design, and the emotional response to the *Attention Needed* framing, the *calling goal* construct, and the *History* screen in particular — are reported and discussed in Chapter 4.

Chapter 4: Results and Evaluation

This chapter reports findings from the prototype evaluation (§3.6) and assesses *keepclose* against the requirements (§3.3) and design hypothesis (§3.5).

4.1 Chapter overview

The evaluation produced three streams of data: task-completion observations against the checklist defined in §3.6.4; paraphrased and selectively quoted answers to the post-task and overall open-ended questions; and the researcher's reflection notes recorded immediately after each session. The chapter draws on all three in combination — task-completion data anchors how the prototype performed at the level of individual interactions, while the open-ended responses and reflection notes interpret what those interactions meant to participants.

§4.2 reports task-completion findings across ten participants and three scenarios. §4.3 presents seven themes from the open-ended responses, organised thematically because the strongest findings cut across more than one scenario. §4.4 evaluates the prototype against the functional and non-functional requirements (§3.3), distinguishing those for which the evaluation provided direct evidence from those for which the evidence is indirect or which the method could not test. §4.5 discusses what the findings collectively say about the design hypothesis and addresses the evaluation method's limitations, before §4.6 closes the chapter. Direct verbatim quotes are marked with ★ to distinguish them from paraphrased session-note material; all participant material is anonymised by participant code (P01–P10), in line with the consent terms described in §3.7.

4.2 Task completion

The first data stream is the task-completion record against the checklist of expected sub-steps for each scenario. For each participant and sub-step, the outcome was recorded as *Completed*, *Partial / hint required*, or *Failed*. Across ten participants, three scenarios, and five sub-steps per scenario — 150 observations — there were no outright failures: the interaction model carried participants through all three journeys without anyone reaching a dead end.

Three design issues produced the partial completions, each described below. Table 4.1 reports counts.

Scenario	Sub-step	Completed	Partial / hint	Failed
S1: Drift to Call	Found Mum in *Attention Needed* section.	10	0	0
	Tapped Mum's card to open Contact Detail.	10	0	0
	Noticed and used the time-zone information.	8	2	0

Scenario	Sub-step	Completed	Partial / hint	Failed
	Tapped *Call now* and saw the launch sheet.	9	1	0
	Indicated they would launch a call.	10	0	0
S2: Add a Contact	Identified and tapped the floating action button.	9	1	0
	Reached the Add Contact form.	10	0	0
	Tapped a field and reached filled state.	10	0	0
	Tapped *Save*.	10	0	0
	Returned to Contacts with Sara added.	10	0	0
S3: Review Patterns	Found the *History* tab in bottom navigation.	10	0	0
	Reached the History screen.	10	0	0
	Understood the mini-strip pattern.	5	5	0
	Noticed 4-week / 8-week range toggle.	7	3	0
	Indicated they had finished reviewing.	10	0	0

Table 4.1 — Task-completion counts across ten participants and three scenarios

The three design issues that produced these partial completions each reflect a design choice rather than participant-specific difficulty, and each is worth describing in detail.

The first is the time-zone helper in Scenario 1 (8/10 noticed and used). The two participants scored partial — P02 (8-hour difference) and P08 (4-hour difference) — bypassed the helper for different reasons. P08 focused on emotional and privacy implications, paying less attention to the time-zone cue; P02 noticed the helper but read it as background information rather than a decision input. This is picked up in §4.3.1.

The second is the Quick Call Launch sheet in Scenario 1 (9/10 reached without hint). P10 needed a hint, having expected *Call now* to launch a call directly rather than present a chooser. After seeing the picker they understood the pattern and suggested choosing a default app during contact setup — less a usability failure than an expectation mismatch, returned to in §4.3.6.

The third and most consequential is the History screen in Scenario 3. Five of ten participants needed a moment to infer what the four-tile mini-strip represented (one tile per week, ticked if a call took place); three did not notice the 4-week / 8-week range toggle until looking further around the screen. Both findings indicate a missing legend on the History screen — the most concrete usability finding the evaluation produced — and both were flagged spontaneously: P01 suggested a tiny legend on *History*, and P05 wanted a simple explanation of the history squares. The design implication is discussed in §4.3.5.

The Add Contact journey (Scenario 2) was completed without issue by nine of ten participants. The only partial case was P05, who needed a hint to notice the floating action button on Contacts. This suggests a minor discoverability issue on first encounter with the app, but because it affected only one participant and did not block task completion overall, it was not pursued further.

The interaction model carried all ten participants through all three journeys, and the task-completion data isolates two narrowly-scoped design issues (the time-zone helper's variable utility, and the missing legend on the History screen) and one expectation mismatch (the call launch sheet). The substantive findings, however, lie not in task completion — which the prototype handled well — but in the open-ended responses reported next.

4.3 Themes from open-ended responses

Analysis of the open-ended responses and reflection notes surfaced seven themes recurring across multiple participants and scenarios. The themes are presented in the order they bear on the design, from those participants endorsed to those that surfaced concerns. Each is supported by participant counts and anonymised quotations.

4.3.1 The time-zone helper as practical and emotional value

The time-zone helper was the most consistently and warmly endorsed feature. Eight of ten participants spontaneously commented on it during Scenario 1 or returned to it in the overall questions, and none expressed reservation. The endorsement had two strands. The first was practical: participants described the helper as removing one of the small uncertainties — *am I about to call at a bad time?* — that the user research had identified as a drift contributor. The second, more interesting, was that several participants framed the helper not as logistical assistance but as removing a permission to delay. P07, with the largest time-zone difference in the sample (9 hours), articulated this in a ★-marked verbatim quote: *"The time-zone part removes the excuse of not knowing."* P02 (8 hours) made the same point as a paraphrase — that the biggest barrier was uncertainty about the time on the other end, and that the helper would prompt earlier calls rather than postponement. The helper made the time difference visible in a way that left no room for the reflexive *"I'll call later when I'm sure it's a good time"* — a small but real anti-procrastination move.

4.3.2 The *Attention Needed* framing — useful surfacing, vulnerable wording

The *Attention Needed* heading on Contacts produced the most divided response of any element in the prototype, and is the clearest example of the asymmetry between the value of a feature and the cost of its wording (§3.5.3). All ten participants understood what the heading was doing — surfacing contacts whose call frequency had fallen below the user's stated goal

— and at the level of *function*, the response was overwhelmingly positive. P05 smiled when seeing Mum at the top of the section, observing that the experience resembles what happens during busy weeks; P01 articulated the value in a ★-marked verbatim quote: *"It would make the call come back into my mind before it becomes awkward."*

The wording, however, was a different matter. Eight of ten participants raised concerns; only P03 and P09 found it acceptable. The objections clustered into two registers. Milder concerns were about formality and tone: P05 felt *"attention"* was a little formal for family, P01 found the wording slightly serious, and P07 wanted it to feel less urgent. Stronger objections concentrated on the heading reading as obligation or judgement, captured in three ★-marked verbatim quotes: P02's *"I already know I should call; I don't want the app to sound like my conscience"*, P04's *"I want help remembering, not a small alarm telling me I failed my family"*, and P08's *"Relationships are not overdue library books."* P06 disliked the phrase because it sounded like a task-management app, and P10 said the wording felt too much like a problem notification — adding the ★-quoted recommendation that the design *"sound like care, not a warning."*

This confirms the design risk flagged in §3.5.3, where *Attention Needed* was identified as a borderline microcopy choice needing evaluation feedback. The mechanism of difficulty — the wording, not the function — points to the direction of the fix.

4.3.3 Home/Local Circles — workable but binary

The Home / Local Circle structure was the prototype's second most-discussed element, and the response mirrored the R15 finding in the user research that motivated NFR7's editable-labels requirement (§3.5.3). Six of ten participants pushed back against the binary in some form; a smaller group (P01, P05, P10) found the default structure straightforwardly useful, with P01 adding that labels should remain editable.

The pushback did not split cleanly by time abroad. Two participants at early stages (P02, 6 months; P07, 5 months) wanted more flexibility alongside three at later stages (P04, 3 years; P06, 4 years; P08, 2.5 years); P09 (1.5 years) also wanted additional categories. P06 captured the underlying issue in their ★-marked verbatim quote: *"My home is not only where my passport is."* P08's notes record that real relationships felt *messier than Home/Local*, while accepting a minimal structure is needed for the app to function. P04 raised the related concern that the binary could flatten complex relationships, especially for students with family in multiple countries.

A related pattern was the desire for additional or custom Circles. P02's notes record specific suggestions (Family, Close friends, UK friends), and P09 independently raised the same idea. Two participants made the editability request in their overall feedback: P06 recommended Circles be editable from first setup, and P01 recommended a clearer way to edit Home/Local labels. The implication is that the editable-labels requirement (FR7) needs to be foregrounded earlier than the prototype currently does — implied as a Settings-area capability but not exposed in the Add Contact form, where participants would benefit from editing or renaming Circles at the moment they first encounter them. This is an actionable finding for future work.

4.3.4 Calling goals — useful structure, fragile language

Nine of ten participants endorsed the *Calling Goal* construct on the Add Contact form; only P08 said goals did not fully resonate. The pattern was that goals provide structure where everyday life erodes intent. P10's notes record that *"without a goal, weeks can pass quickly"*, with the per-contact framing (rather than a single global setting) a key part of the appeal. P01 described the weekly goal as a gentle intention rather than a strict target, qualified by the condition that the app does not punish missed weeks. P05 endorsed goals strongly because *once a week* made a vague responsibility feel manageable, while stressing they should remain adjustable. P07 added that goals matter precisely because when someone is far away, "call sometime" becomes too vague.

The endorsement was qualified in two ways. First, participants made a contact-type distinction: goals felt right for parents and siblings but inappropriate for friendships. P02's notes record that the goal idea worked for practical relationships, but for friends a formal goal could make the relationship feel scheduled or less spontaneous. P06 wanted a *"no goal"* option for relationships where frequency varies naturally. P09 added that goals were unnecessary where routine already exists (e.g. parents) but useful for people they care about but forget to contact.

Second, the word *goal* itself drew pushback. P04 preferred *"ideal rhythm"*; P03 preferred *"soft intentions"*; P08 preferred a *"remind me if it has been a while"* approach over a fixed frequency. The directiveness was felt to misrepresent the relational nature of the underlying intention.

Both qualifications point in the same direction: the structural decision is sound, but the language wrapping it is too directive. The per-contact frequency selector could be reframed by changing the label from *Calling Goal* to a softer formulation, with no change to the underlying interaction model. This is a microcopy fix rather than a design rethink, and like the *Attention Needed* finding it can be acted on at low cost.

4.3.5 History as calm reflection, with a desire for depth

The History screen produced the most varied emotional responses of any element in the prototype, and is the screen for which the evaluation provides both the strongest endorsement of a design decision and the strongest evidence that the decision created secondary problems.

The endorsement first: across all ten participants, the deliberate omission of streaks, scores, and rankings (NFR2) was unanimously supported. P03 commented that scores would make it feel *"like Duolingo"*; P06 said the screen looked like information, not judgement; P01 described it as feeling more like a reflection tool than a productivity tracker. Several participants explicitly contrasted the History screen with the design language of fitness and productivity apps and endorsed the choice.

The emotional response varied markedly. Four participants (P01, P03, P06, P09) reported feeling calm or neutral; two (P05, P07) curious and mildly comforted; three (P02, P04, P10) a mild guilt or reflectiveness when seeing grey weeks for family contacts; and P08, the most articulate on this point, reported feeling judged/guilty, even though the design was visually calm, with their notes recording that grey gaps *still looked like evidence of failure* and that they would want the option to hide the History screen entirely. P08 added that concrete

feedback would make the guilt worse for them. This split is itself a design finding: calm visual language reduces but does not eliminate the emotional weight of seeing one's own pattern of contact. A wellbeing-supportive design needs an opt-out path for users who experience the reflective surface negatively — a need NFR7 (§3.3.3) anticipated at the level of principle, but which the evaluation indicates needs to be designed into the History screen specifically.

The other strand was a request for more depth — but *qualitative* depth, not numeric. Six participants suggested small additions: P07 wanted to tap each square to see what happened that week (progressive disclosure: simple first, details if tapped); P01 wanted a small legend confirming that each square represents a week; P10 suggested a short explanation and a gentle monthly reflection; P02 and P05 both wanted more textual feedback; and P04 wanted a gentle sentence reminding users that patterns vary. No participant asked for streaks, scores, percentages, or aggregate metrics. The request is for context, not measurement — a distinction with real design implications, since qualitative depth strengthens the reflective frame, while numeric depth would convert the screen into the kind of dashboard NFR2 forbids. The legend issue, which also surfaced in the task-completion data (§4.2), is the most concrete fix.

4.3.6 Layer-not-replacement, and the regional apps gap

The decision to design *keepclose* as a layer over existing calling apps rather than a replacement was endorsed by all ten participants. The endorsement was often framed in terms of what the design avoids rather than what it offers: P06 said existing apps already contain their social context and trust; P01 would not want a new calling platform; P03 would not trust or need a new video-calling feature; and P08 identified the layered approach as *safer than replacing* existing tools. P09 articulated the prototype's intended value most precisely: the value is not the call itself, but deciding who to contact and then quickly launching the right app.

Two narrower findings emerged. The first is an expectation mismatch on the launch sheet: P10 had expected *Call now* to launch a call directly rather than present a chooser, and the resulting hesitation was scored as a partial completion (§4.2). After the picker appeared, P10 understood it and suggested choosing a default app during contact setup — a request echoed by P03, who wanted a faster “use default app” option after first setup.

The second is what might be called the *regional apps gap*. P07, with a 9-hour time difference, said the picker should include maybe *WeChat* depending on the person, naming non-Western apps the prototype's default chip set does not surface. Although the finding rests on a single participant, it points at a real-world implementation requirement: a tool for international students cannot ship with a calling-app picker that assumes WhatsApp and FaceTime are universal. The *"Other"* affordance is necessary but not sufficient.

4.3.7 Privacy and the spectre of surveillance

Although the prototype was deliberately designed to avoid automatic call-history access (NFR3) and to keep contact data conceptually local (NFR6), three participants raised privacy concerns spontaneously. The pattern is itself the finding: participants did not object to the design's actual data behaviour — manual logging, no automatic call-history access — but

expressed unease that an app designed around relationships *could* drift toward surveillance, even if this prototype did not.

P08 expressed this most strongly, raising it at several points. Their notes record that they would want clear reassurance that the app does not read WhatsApp, contacts, or call logs automatically; that they questioned whether the app would make them feel monitored; and that if the app felt like surveillance of relationships, they would uninstall it. P08 is the same participant who was most critical of the History screen in §4.3.5 — the two findings together suggest a single participant whose wellbeing-protection concerns crossed both privacy and emotional-surface domains.

P04 made the related suggestion — not as personal anxiety but as a design recommendation — that the app should include a privacy note explaining that call history is manually logged; the researcher reflection note for that session identified privacy transparency as something to be made visible earlier. The implication is not that NFR3 or NFR6 need tightening — the prototype already meets the substantive concern — but that the *reassurance* needs to be surfaced where users will see it. The principle is sound; its visibility is what the evaluation indicates is missing. This finding concerns trust signalling rather than interaction design, pointing to a refinement of the onboarding experience rather than any change to existing screens.

A related concern surfaced in P02's session: their notes record the worry that the app could start feeling like emotional homework if reminders became too frequent. This is the same fear as P08's, expressed differently — not of data leaving the device but of the relationship itself becoming a measured object. NFR7's opt-in feature depth is the design response, and the evaluation gives it qualitative support: feature depth needs to be opt-in not because some users will not need it but because some users will actively resist it.

4.4 Evaluation against requirements

This section examines each functional and non-functional requirement (§3.3) against the evidence the evaluation produced. Three categories are used. *Directly tested* indicates that the procedure exercised the requirement through one of the three scenarios. *Indirectly evidenced* indicates that the requirement was not the focus of any scenario but produced relevant participant comment, typically through the post-task or overall questions. *Not testable in this format* indicates a requirement whose effect can only be assessed under conditions the evaluation could not provide — typically extended use over time, or interactions the prototype was not built to simulate. The third category matters because some requirements — particularly those concerning behaviour over time — cannot be assessed in a single 30-minute session, and naming this avoids overstating what the evaluation can verify.

4.4.1 Functional requirements

Table 4.2 maps each functional requirement onto evaluation evidence and a verdict.

ID	Requirement	Evidence type	Evidence summary	Verdict
FR1	Per-contact frequency goal	Directly tested (S2)	All ten participants set a calling goal during Add Contact; nine endorsed the construct in the post-task questions, with a microcopy concern about the word *goal* (§4.3.4).	Met; refinement needed in label wording.
FR2	Time-zone-aware suggested call windows	Directly tested (S1)	Eight of ten noticed and used the helper without prompt; two partial completions came from participants whose time-zone use of the helper was unnecessary or incidental in their case (§4.2, §4.3.1). The helper was the most consistently endorsed feature.	Met; prominence could appropriately scale with time-zone difference.
FR3	Gentle reminders surfacing drift	Directly tested via the surfacing element (*Attention Needed*) on Contacts (S1)	All ten participants understood and endorsed the surfacing function; eight raised concerns about the wording (§4.3.2). The notification mechanism itself was not exercised — the *Attention Needed* heading was tested as a static, on-screen indicator.	Met at the level of in-app surfacing; reminder *delivery* (push notifications) is not testable in this format.
FR4	One-tap launch into preferred calling app	Directly tested (S1)	Nine of ten reached the launch sheet without hint; the layered model was endorsed by all ten (§4.3.6). One expectation mismatch (P10) suggested an opportunity for default-app shortcuts.	Met; refinement to support a saved default during contact set-up.
FR5	Optional outcome logging	Indirectly evidenced (S1)	The *Log a call instead* affordance was visible on Contact Detail during S1 but no scenario asked participants to use it. Participants who commented on logging endorsed its optional framing; no participant requested mandatory or automatic logging.	Met at design-level; not exercised behaviourally.

ID	Requirement	Evidence type	Evidence summary	Verdict
FR6	Calm, non-comparative history view	Directly tested (S3)	All ten reached the History screen; five needed a moment to interpret the mini-strip and three did not initially notice the 4w/8w toggle (§4.2). The non-comparative framing was unanimously endorsed (§4.3.5).	Met in framing; clarity of the mini-strip pattern needs a legend.
FR7	Editable contact circles with Home/Local default	Indirectly evidenced (S2)	The Home/Local default was visible during Add Contact, but circle editing post-creation was not in the prototype. Six participants raised the binary as inadequate, with the pushback spread across both early and later stages of international study (§4.3.3); custom-label requests were spontaneous.	Met for the default; the editability promised by the requirement is not realised in the prototype, and the evaluation indicates editing should be foregrounded earlier in the user journey.

Table 4.2 — Functional requirements against evaluation evidence

The most important entry is FR7. The requirement specifies editable circle labels, and the evaluation provides strong evidence that editability matters — but the prototype does not implement label editing. Had the prototype exposed circle editing within Add Contact, several of the §4.3.3 concerns might have been resolved within the session. This is acted on in the conclusions chapter as a priority for future work.

A related limitation concerns FR3. The requirement is for *gentle reminders*, which in a real product would be delivered as push notifications. The prototype exercises only the in-app surfacing element (the *Attention Needed* heading), not notification delivery itself. The *Attention Needed* findings tell us how the surfacing reads on screen; they tell us nothing about how the same wording would land as a notification. This is a limitation of what the format can test, not of the requirement.

4.4.2 Non-functional requirements

Table 4.3 maps each non-functional requirement onto evaluation evidence.

ID	Requirement	Evidence type	Evidence summary	Verdict
NFR1	Configurable, conservative reminder frequency	Not testable in this format	Reminder frequency cannot be assessed in a single 30-minute session, since the relevant failure mode (reminder fatigue) emerges over weeks. The *intent* of the requirement is supported by the §4.3.2 microcopy findings.	Not testable; principle endorsed by participant pushback against pressure-based wording.
NFR2	No streaks, scores, or gamification	Directly tested (S3)	Unanimously endorsed across all ten participants. Several explicitly contrasted the History screen with productivity and fitness apps and said the absence of metrics was the right design call (§4.3.5).	Met; this is the strongest single endorsement in the evaluation.
NFR3	No automatic call-log access	Indirectly evidenced	The prototype contains no automatic call-log access; participants raised privacy concerns spontaneously and one participant explicitly recommended surfacing a privacy note explaining the manual logging model (§4.3.7).	Met; visibility of the principle to users is the gap.
NFR4	Gentle, neutral, or encouraging copy	Directly tested across all three scenarios	The mixed reception of *Attention Needed* (§4.3.2) and *Calling Goal* (§4.3.4) shows that the requirement was met in some places and missed in others. The principle was endorsed; specific copy choices need rework.	Partially met; specific microcopy needs revision.
NFR5	Two-tap minimum interaction path	Directly tested (S1)	The Contacts → Contact Detail → Call now path was completed by nine of ten without hint; reaching the launch sheet takes 2 taps from Contacts, and the third tap selects the calling app (§4.2, §4.3.6).	Met for in-app navigation; the third tap to choose a calling app would be removed by the saved-default refinement identified for FR4.

ID	Requirement	Evidence type	Evidence summary	Verdict
NFR6	Local-first contact data	Not directly testable	A Figma prototype does not transmit data; the requirement's intent is reinforced by the §4.3.7 privacy findings, but the evaluation cannot test the implementation.	Stated principle, indirectly endorsed.
NFR7	Opt-in feature depth	Indirectly evidenced	P08's request for a way to hide the History screen and the broader concern about *"emotional homework"* (§4.3.5, §4.3.7) directly support this requirement. The prototype itself does not implement an opt-out path.	Principle endorsed by evaluation; implementation needed.

Table 4.3 — Non-functional requirements against evaluation evidence

Three patterns emerge across the two tables. Requirements describing design *function* (FR1, FR2, FR4, FR6, NFR2, NFR5) were generally well-served by the evaluation: scenario walkthroughs are a good fit for testing what a design *does*. Requirements describing design *over time* (NFR1 reminder frequency, NFR6 data behaviour) are not testable in a single-session format and were inherited as principles rather than verified findings. Requirements describing design *protection from* something (NFR2 against gamification, NFR3 against surveillance, NFR7 against forced engagement) produced strong endorsement once participants encountered the design, but in two cases (NFR3 visibility, NFR7 opt-out path) the principle is in the requirements without being in the prototype. The next section discusses what the findings, taken together, mean for the project's central design hypothesis.

4.5 Discussion and limitations of the evaluation

The evaluation set out to answer two questions (§3.6.1): whether international students could complete the prototype's three core journeys with low cognitive effort, and whether the prototype *felt* the way the design intended — calm rather than judgmental, supportive rather than surveilling, autonomy-respecting rather than obligation-creating. The findings (§4.2, §4.3) answer both, with different degrees of confidence and different implications for the design hypothesis.

What the evaluation says about the design hypothesis

The project's central design hypothesis (Chapter 1, §3.3, §3.5) was that a low-friction, calm, autonomy-respecting calling layer — one that surfaces drift without instrumenting relationships, reduces the friction of acting on intent, and refuses the gamified vocabulary of productivity apps — could plausibly support international students in maintaining meaningful contact. The evaluation provides qualified support, strongest where the design's restraint is most visible.

The clearest endorsement is for what the design *does not* do: the unanimous support for the absence of streaks, scores, and rankings (NFR2, §4.3.5); the endorsement of the layered-not-replacement model (§4.3.6); the explicit contrast participants drew between the History screen and productivity-app vocabulary; and the privacy concerns that arose around what the design *might* drift toward rather than what it actually does (§4.3.7). Participants recognised the design was operating in a different register from the productivity and habit-tracking tools they associated with this kind of feature, and endorsed the choice — reaching the design hypothesis's conclusion themselves without being told it.

The next-strongest endorsement is for the design's pragmatic choices: the time-zone helper (§4.3.1), the two-tap call path (§4.3.6), and the per-contact frequency goal (§4.3.4). These respond most directly to the drift-cause findings (Q14, §3.2.3), and the evaluation corroborates the chain from research to design to user response. The participant who described the helper as removing *"the excuse of not knowing"* (P07) put it more sharply than the design rationale: the helper's value is anti-procrastinative, not just informational.

The qualifications cluster around two issues. The first is microcopy: *Attention Needed* (§4.3.2) and *Calling Goal* (§4.3.4) both carry more directive weight than the design intended, producing participant-suggested replacements around warmer, less task-oriented framings — confirming the risk §3.5.3 anticipated and identifying the direction of the fix. The second is design *visibility*: privacy (NFR3) and opt-out paths on the History screen (NFR7) need to be visible to the user, not merely present in the implementation. Both point at refinements rather than fundamental redirection.

The honest reading is that the design hypothesis is supported but not proven. The prototype does the things it was designed to do, in the register it was designed to do them; the evidence is consistent with the proposition that this kind of design could plausibly support wellbeing-oriented contact maintenance. The further claim — that it *would* support wellbeing in actual use over time — is one the evaluation method cannot make, and the next sub-section explains why.

Limitations of the evaluation

The limitations anticipated in §3.2.4 and §3.6.7 can now be revisited with reference to what was and was not learned. The single-session format constrains what the evaluation can claim about wellbeing outcomes: the project's hypothesis is fundamentally about behaviour and emotion over time, but a 30-minute walkthrough cannot test this directly. Participants' reports of how the design feels and how they imagine using it are valuable evidence for how the design lands at first contact, but weaker evidence for the questions the hypothesis ultimately needs answered — whether the time-zone helper retains its value with repeated use, whether the *Attention Needed* surfacing remains welcome rather than becoming a source of frustration, and whether the History screen rewards or burdens occasional checking.

A second cluster of limitations concerns the artificiality of both scenarios and sample. Participants imagining a call to their own mother are not in the same emotional state as someone who has not actually called their mother in weeks. This is most consequential for the §4.3.5 History findings: participants who reported guilt at seeing grey weeks were responding to a hypothetical pattern of contact. The direction of the finding is likely correct, but the magnitude in real use cannot be inferred from session evidence. Similarly, the §4.3.2 *Attention Needed* findings tell us how the wording reads in a session, not how it would read

as a real notification after a real period of no contact. The sample compounds this: ten participants is within the typical 6–10 range for this kind of project, and diversity across time abroad, time-zone difference, course level, and platform (§3.6.3) was deliberately broad — but ten participants is too few for fine-grained subgroup analysis, and self-selection limits what can be claimed about students who would not have volunteered.

Some requirements were inherently outside what the format could test: NFR1 (reminder frequency, requiring temporal exposure), NFR6 (data behaviour, requiring an implementation), and FR3's notification-delivery aspect (§4.4). These are boundaries of single-session prototype evaluation rather than weaknesses of the method, but they qualify the verdicts in Tables 4.2 and 4.3. Finally, the decision not to run a second iteration (§3.6.7) limits what this chapter can report: the findings describe the prototype as initially evaluated, and the design implications — softer microcopy, foregrounded label editing, a History-screen legend, an opt-out path, default-app shortcuts, surfaced privacy reassurance — are described in Chapter 5 as future work rather than evaluated as resolved.

The evaluation produces strong evidence about the design's first-contact plausibility and weaker evidence about its sustained-use wellbeing impact. The findings are best understood as supporting the design's direction and identifying where its refinements should land, rather than as a verdict on whether the design would deliver the wellbeing benefits it is intended to support in real-world use. That verdict would require longer-term real-world evaluation, discussed as future work in Chapter 5.

4.6 Summary

This chapter has reported findings from the prototype evaluation (§3.6) and assessed *keepclose* against the requirements (§3.3) and design hypothesis (§3.5). Task-completion data (§4.2) indicated the interaction model carried all ten participants through the three core journeys without outright failures, and isolated three narrowly-scoped design issues — the variable utility of the time-zone helper, the missing legend on the History screen, and an expectation mismatch on the call-launch sheet. Seven themes from the open-ended responses (§4.3) collectively endorsed the design's pragmatic choices and its restraint in avoiding gamified vocabulary, while identifying microcopy on *Attention Needed* and *Calling Goal*, the binary Home/Local Circle structure, and privacy-reassurance surfacing as the principal areas needing refinement. The requirements assessment (§4.4) confirmed that requirements describing design *function* were well-served by the evaluation, while requirements describing design *over time* and design *protection from* something were either outside the format's reach or partially realised. The discussion (§4.5) concluded that the evaluation provides qualified support for the design hypothesis at the level of first-contact plausibility, acknowledging that the sustained-use wellbeing impact lies beyond what a single-session walkthrough can verify.

The implications of these findings for the *keepclose* design, the project's contribution, and future work are drawn out in Chapter 5.

Chapter 5: Conclusions

5.1 Chapter overview

This chapter assesses the extent to which the project's original aim, hypothesis, and objectives were met (§5.2 and §5.3), draws together what was learned about the design output and about the methods employed (§5.4), and identifies directions for future work that the project's findings imply (§5.5). The chapter closes with a short reflection on the project's contribution and on the design process through which it developed (§5.6).

5.2 Assessment of objectives

The five objectives stated in §1.3 were addressed to varying extents, with one substantive scope change (the iteration cycle) noted in the introduction and revisited below.

O1 — literature review. Met. Chapter 2 covers homesickness and loneliness in international students, the use of digital technologies for sustaining transnational connection, the broader HCI tradition of awareness and lightweight presence, and the wellbeing-supportive design framework adopted as the project's design lens. The literature review explicitly responded to the breadth concern raised in the project proposal feedback by extending beyond student-loneliness work into adjacent literatures on technology-mediated connection.

O2 — user research. Met. The 50-respondent survey (§3.2) surfaced clear evidence about connection drift, the value of voice/video calling specifically, and the features and deterrents that would shape the prototype's design. The findings were sufficient to inform the requirements derived in §3.3 without supplementary interviews, though the survey-only approach is itself acknowledged as a limitation in §3.2.4.

O3 — requirements and information architecture. Met. The seven functional and seven non-functional requirements (§3.3) and the five-area information architecture (§3.4) were derived from the user research findings with explicit traceability between each requirement and its survey source. The requirements served their dual purpose: they shaped the prototype, and they provided the structure against which the evaluation in §4.4 assessed the design.

O4 — prototype build. Met. The high-fidelity Figma prototype (§3.5) consists of seven screens supporting three end-to-end user journeys, with full visual and tonal design rather than wireframe-level fidelity — a choice the design hypothesis itself required, since tone and visual restraint cannot be evaluated at low fidelity.

O5 — evaluation. Substantially met, with one scope change. The scenario-based walkthroughs with ten international students (§3.6) produced the qualitative evidence Chapter 4 reports against. The proposal's planned post-evaluation iteration was, however, scoped as optional during the project, and not carried out; the analytical depth of the single round of findings was prioritised over a partially completed second iteration. The implications of this scope change are discussed in §5.5 as future work.

5.3 Assessment of the design hypothesis

The design hypothesis articulated in §1.2 — that a low-friction, calm, autonomy-respecting calling layer can plausibly support international students in maintaining wellbeing-oriented contact with the people who matter to them — received qualified support from the evaluation reported in Chapter 4.

Where the evaluation evidence was strongest was in support of what the design *does not* do. Participants unanimously endorsed the deliberate omission of streaks, scores, and gamified vocabulary (§4.3.5); they spontaneously endorsed the layered-not-replacement model (§4.3.6); and the privacy concerns they raised concerned what an app of this kind *could* drift toward rather than what the prototype actually does (§4.3.7). The participants reached the design hypothesis's conclusion themselves: they recognised that the design was operating in a different register from the productivity and habit-tracking tools they associated with this kind of feature, and they endorsed the choice.

The qualifications cluster around two issues. The first is microcopy: the *Attention Needed* and *Calling Goal* phrasings carry more directive weight than the design intended (§4.3.2, §4.3.4), confirming the risk §3.5.3 had anticipated. The second is design *visibility*: the prototype's protective principles around privacy (NFR3) and opt-out paths (NFR7) need to be visible to the user, not merely present in the implementation (§4.3.7, §4.4). Both qualifications point at refinements rather than fundamental redirection.

The honest reading is that the design hypothesis is supported but not yet proven. The prototype does the things it was designed to do, in the register it was designed to do them; the evaluation evidence is consistent with the proposition that this kind of design could plausibly support wellbeing-oriented contact maintenance. The further claim — that it would support wellbeing in actual sustained use — is one the single-session evaluation method cannot make, and is identified in §5.5 as future work requiring longer-term, real-world evaluation.

5.4 What was learned

About the design

The strongest substantive learning from the project concerns the value of *deliberate restraint* in wellbeing-oriented relational technology. The most reliably endorsed elements of *keepclose* were not the features the prototype contained but the features it deliberately omitted: no streaks, no scores, no aggregate connection-health metric, no gamified comparison between contacts. Participants in the evaluation recognised this restraint, and several explicitly contrasted the design with the productivity-app vocabulary they associated with similar features in other contexts (§4.3.5). The implication, beyond the specifics of *keepclose* itself, is that wellbeing-oriented relational technology may need to be defined as much by what it refuses to do as by what it positively offers — a finding consistent with Peters' (2023) framework, applied here to the specific context of international student wellbeing. A second design lesson concerns the gap between a design's protective *principles* and the *visibility* of those principles to the user. The prototype's privacy and opt-out behaviours were sound at the implementation level but underserved at the surface; the evaluation indicates that wellbeing-supportive design needs not only to *be* protective but to

appear protective, particularly in domains where users carry latent concerns about surveillance and obligation.

About the methods

Three methodological lessons emerged from the project. First, the survey-only user research approach was sufficient for shaping a defensible set of requirements at the granularity the design needed; the absence of supplementary interviews was not, in retrospect, a constraint on the design phase, though it limits the qualitative depth of the evidence base in ways acknowledged in §3.2.4. Second, the choice of scenario-based walkthroughs over heuristic evaluation or System-Usability-Scale-style metrics (§3.6.2) was vindicated: the evaluation surfaced both task-level performance and emotional response, and the latter — which the rejected methods would have missed — turned out to be the analytically richer half of the findings. Third, the decision to evaluate at high fidelity rather than low fidelity (§3.5.1) was necessary for the design hypothesis being tested — tone and visual restraint are largely invisible in greyscale wireframes — but limited the kinds of divergent exploration low-fidelity sketching would have enabled at earlier stages. A future iteration of the project, freed from the prototype-already-committed constraint, could productively combine low-fidelity divergent exploration with high-fidelity evaluation.

5.5 Future work

The findings reported in Chapter 4 imply several directions for future development of the design and several more for the next phase of evidence-gathering.

At the level of design refinement, six specific changes follow directly from the evaluation. The microcopy on the *Attention Needed* heading and the *Calling Goal* construct should be replaced with warmer, less directive phrasings drawn from the participant suggestions in §4.3.2 and §4.3.4. Circle label editing (FR7) should be foregrounded earlier in the user journey, exposed within the Add Contact form rather than implied as a Settings-area capability. The History screen needs a small legend or label clarifying the four-tile mini-strip's meaning, and an opt-out path for users who experience the reflective surface negatively (§4.3.5). And the prototype's privacy reassurance — present in the implementation but invisible to the user — should be surfaced explicitly in the onboarding or first-use experience (§4.3.7).

A seventh design direction concerns the *regional-apps gap* identified in §4.3.6. While the prototype's Add Contact form includes an 'Other' affordance for calling apps outside the default WhatsApp/FaceTime chips, this puts the burden on users to surface the apps they actually use. A future iteration should treat the calling-app set as configurable, with non-Western apps (WeChat, and others) surfaced as default chip options when the user's profile suggests they would be relevant.

At the level of evidence-gathering, three larger directions follow. The post-evaluation iteration scoped out of the present project should be carried out: several of the refinements above could be tested in a revised prototype evaluated with a new participant cohort, providing the evidence the present evaluation cannot. Beyond a second iteration, the project's design hypothesis ultimately requires longitudinal real-world evaluation — multi-week deployment with a sample of international students using the prototype on their own phones with their own contacts — to assess whether the calm, autonomy-respecting register holds up

under sustained use rather than only at first contact. And a larger and more systematically diverse participant sample would support the kind of subgroup analysis the present ten-participant evaluation could not (§4.5).

5.6 Closing reflection

The *keepclose* project set out to test whether a particular kind of design — low-friction, calm, autonomy-respecting, refusing the gamified vocabulary of productivity apps — could plausibly support international students in maintaining the relationships that contribute to their wellbeing across distance. The evaluation evidence supports the design's direction at the level of first-contact plausibility, identifies the specific refinements its next iteration should land, and locates the longer-term claim — that this kind of design supports wellbeing in sustained real-world use — as a question for future work. The project's contribution is to the design space of wellbeing-supportive relational technology: it offers an empirically-grounded example of what *deliberate restraint* looks like in a calling-maintenance tool aimed at an emotionally sensitive user group, and provides evidence that the omissions which distinguish such a design from neighbouring product categories are not merely tolerated by users but actively endorsed.

Two reflections close the dissertation. The first concerns the asymmetry between feature and wording: the same design element — reminders for connection — was the most welcome feature when phrased as gentle support and the most-cited deterrent when phrased as obligation. Designs in this space are difficult to evaluate in isolation from the language they wear, and microcopy has emerged from the project as a primary design surface rather than a polish layer. The second concerns the limits of the methods used: a single-session evaluation can establish how a design lands on first encounter, but the questions a wellbeing-supportive design must ultimately answer — whether it remains supportive over weeks of use, and whether it changes the practices it was designed to support — require a different shape of evidence than the present project could produce. The next phase of work on *keepclose*, and on the broader design space the project sits within, is in part defined by the gap between first-encounter findings and sustained-use evidence.

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Appendix A — Survey questionnaire and response distributions

Q1. I confirm that I am an international student.
50 responses

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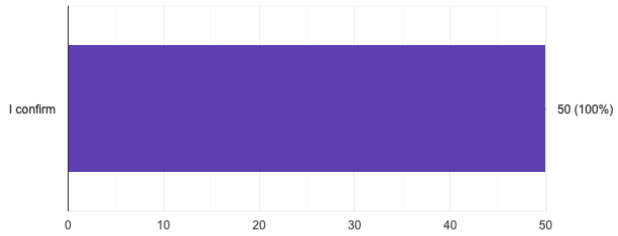

Q2. I confirm that I am aged 18 or above.
50 responses

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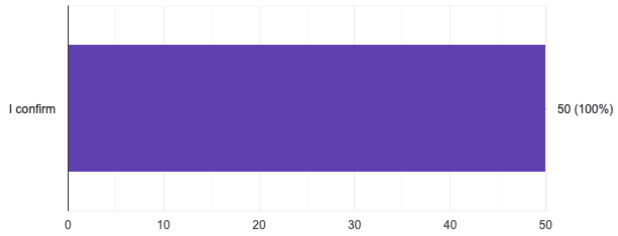

Q3. I agree to take part in this anonymous survey.
50 responses

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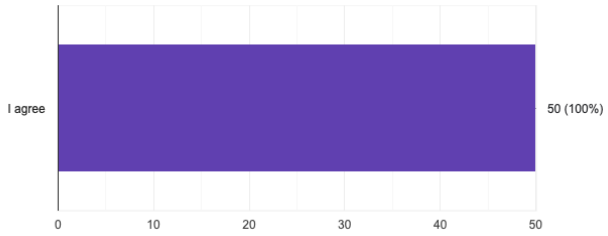

Section 1: About you

Q4. What is your age range?
50 responses

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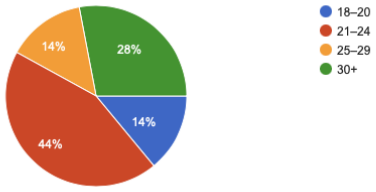

Q5. What level are you currently studying at?

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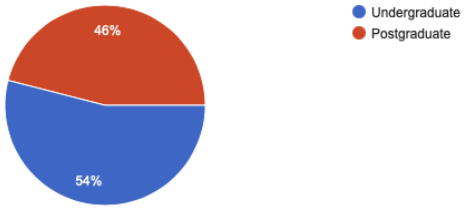
50 responses



Q6. How long have you been living as an international student?

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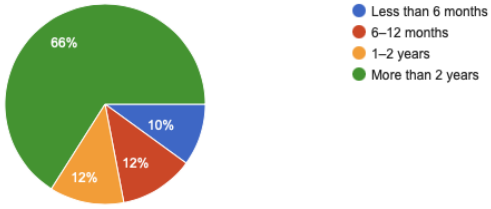
50 responses



Q7. Which best describes your current living situation?

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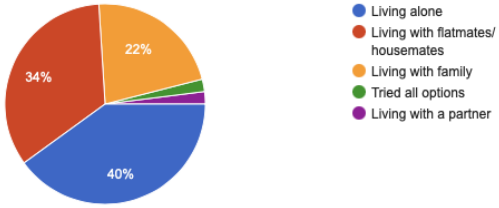
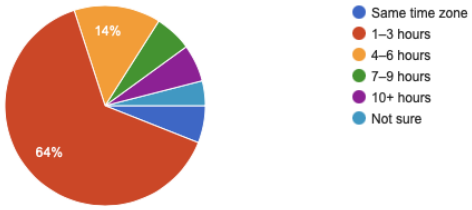
50 responses



Q8. What is the approximate time difference between where you study and your home country?

Copy chart

50 responses

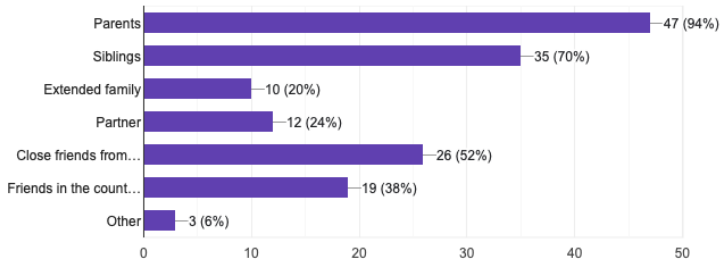


Section 2: How you stay in touch

Q9. Which people do you most regularly keep in touch with while studying abroad? (you can select more than one)

Copy chart

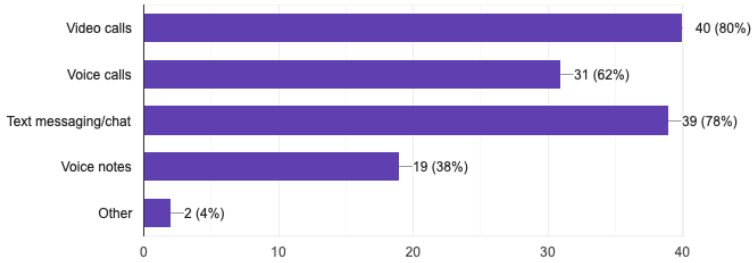
50 responses



Q10. Which communication methods do you use most often to keep in touch with people from home? (you can select more than one)

Copy chart

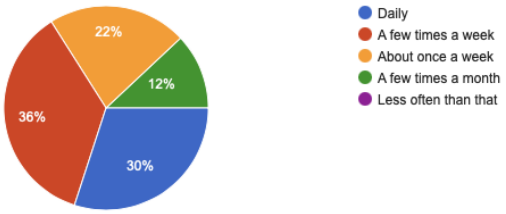
50 responses



Q11. How often do you usually have voice/video calls with people from home?

Copy chart

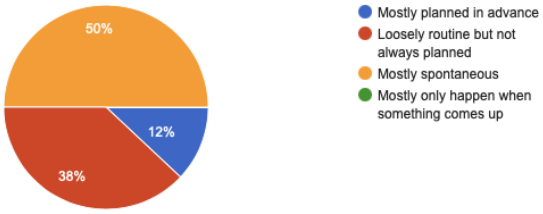
50 responses



Q12. These calls are usually:

Copy chart

50 responses

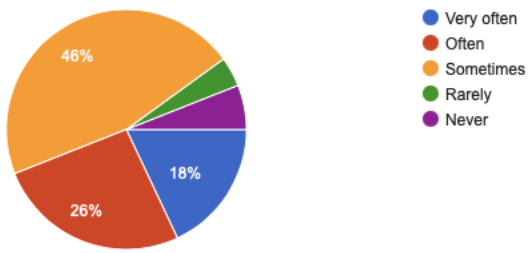


Section 3: When contact starts to drift

Q13. Connection drift: How often do voice/video calls with someone important happen less often than you would like?

Copy chart

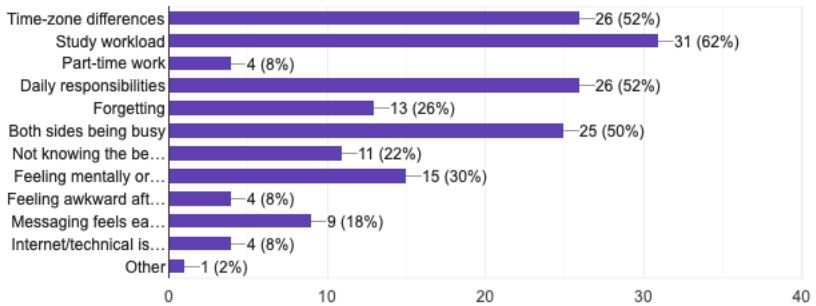
50 responses



Q14. What usually contributes to this connection drift? (you can select more than one)

Copy chart

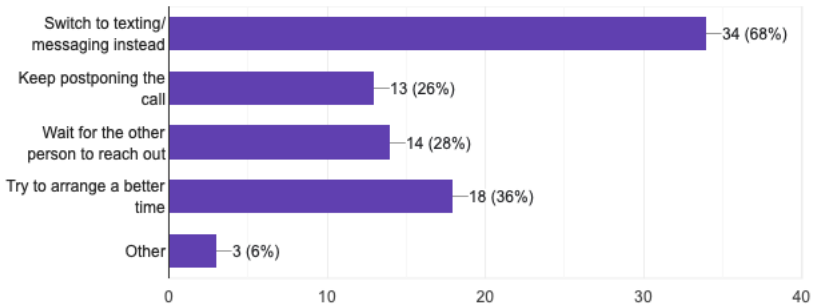
50 responses



Q15. When meaningful calls happen less than you would like, what do you usually do? (you can select more than one)

Copy chart

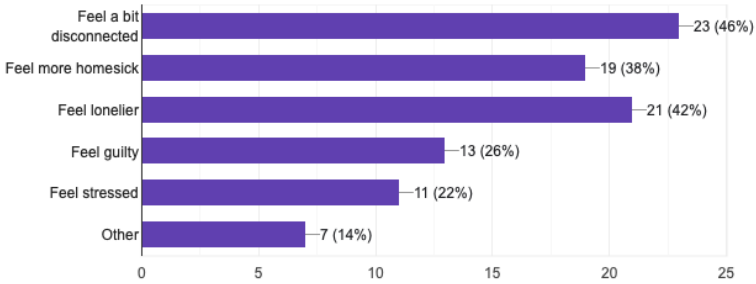
50 responses



Q16. When meaningful calls happen less than you would like, what do you usually feel? (you can select more than one)

Copy chart

50 responses



[OPTIONAL]Q17. In one sentence, as an international student what is the main reason voice/video calls become hard to maintain regularly/routinely?

18 responses

Actually different time with my family.However, I believe 10 minutes calls with my family can get me energetics 🌟

I am busy with other parts of my life, like studying, socialising, worki g or looking for jobs

Both sides busy

My schedule gets too full and I don't know what time of the day to give up.

People drift away, and sometimes you can no longer relate to the problems you both are facing. people choose to talk with people who they share more daily problems and the ones they can create new memories with

Just forget to call when I'm free

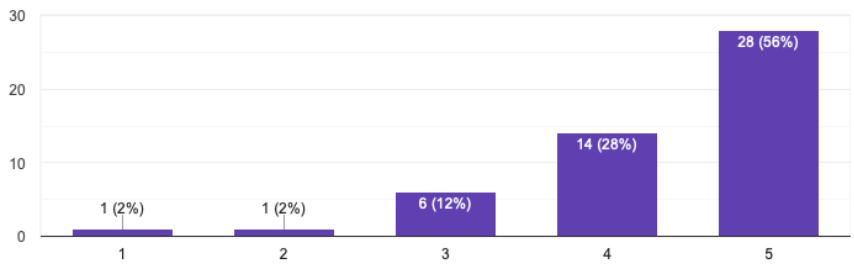
Different expectations as to the frequency of calls from the two parties (can be both insufficiently frequent or too frequent)

Section 4: How important calling feels

Q18. As an international student, how important are voice/video calls for helping you feel connected to people who matter to you?

Copy chart

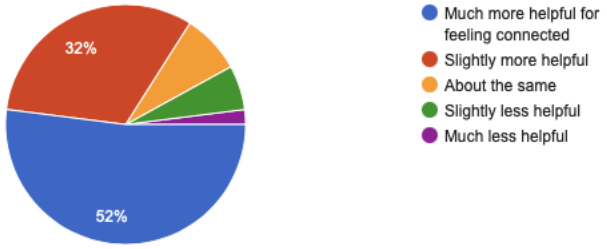
50 responses



Q19. Compared with messaging, meaningful voice/video calls usually feel:

Copy chart

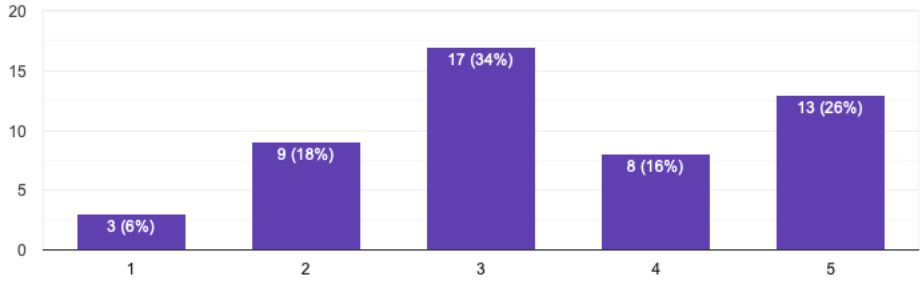
50 responses



Q20. As an international student, to what extent do you agree with the following statement:
"When regular meaningful contact slips for too long, it can negatively affect my sense of connection or wellbeing."

Copy chart

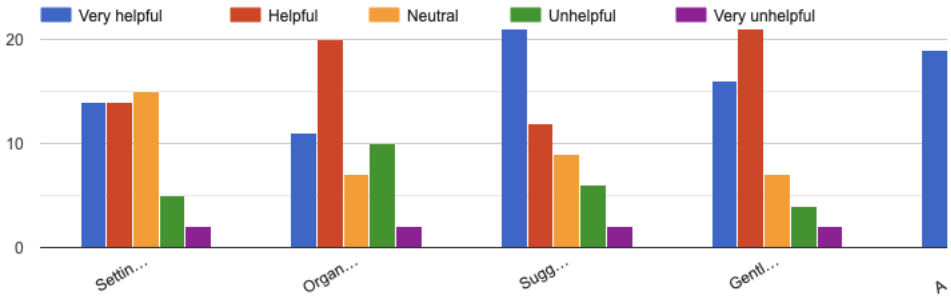
50 responses



[LAST SECTION]Section 5: Possible app features

Q21. For an app designed to help international students maintain regular meaningful calls with friends & family, how helpful would you find the following features:

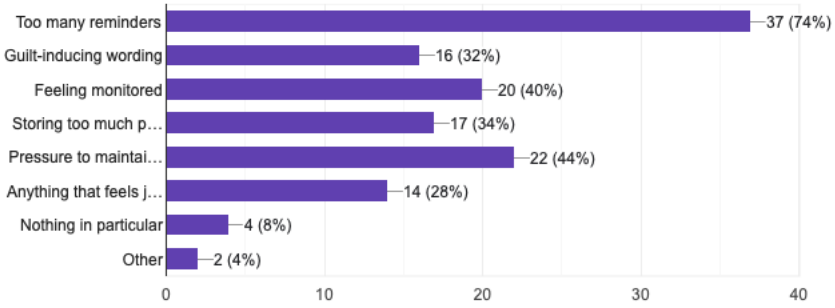
Copy chart



Q22. What might make an app like this less appealing to you? (you can select more than one)

Copy chart

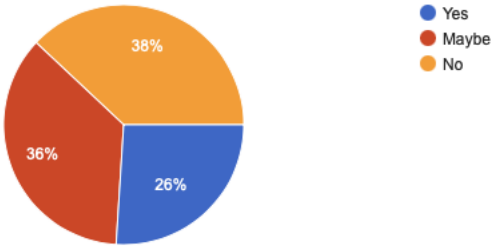
50 responses



Q23. Would you want an app like this to automatically access your call history?

Copy chart

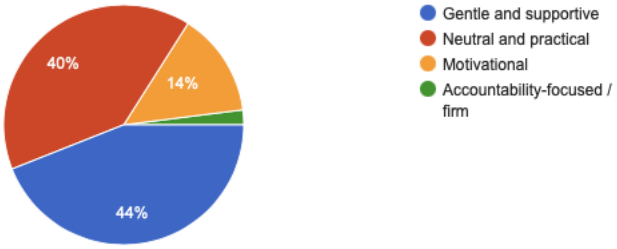
50 responses



Q24. What tone should an app like this have?

Copy chart

50 responses



[Last question - Optional]Q25. For an app designed to help international students maintain regular meaningful calls with friends & family, what suggestions would you have?

8 responses

I don't think i need an app like this. I currently have a once a week call with my parents and that is easy enough to maintain. I dont feel the need to increase this frequency or maintain the same level of relationship with others

Add a lovely, calming romantic ringtone for your close family members.

Some of the things can only add to the internationall student loneliness, like planning calls, setting streaks. If for some reason other person is drifting away, it will only be another reminder of the lost contact. Also putting contacts into "Home" and "Local" can be upsetting, because if the student plans to stay in the country where they studied, this country becomes their home too. When you immigrate, you do not belong to your home anymore, and neither you are fully the citizien of new country. Besides, it feels like people who person meets up with in a new country will have their own app, and might be reluctant to download another one.

I think what can be helpfull is adjustible app UI. If a person wants they can track more stuff, if it harms them(as they only remind about lost contact) they can stop. I also think it can be very helpfull to have some soft reminders from app that it is not easy, and that feeling lonely is normal in the new country. Maybe, app can take data

Appendix B — Scenario script

Study: Supporting Homesickness and Loneliness in International Students — Designing a Wellbeing-Maintenance Calling Tool

Prototype: keepclose

For researcher use during evaluation sessions.

Before I begin

This document contains the scenarios to read aloud and the post-task questions for each of the three core user journeys. Questions are open-ended and focused on the project's aim: whether the prototype could plausibly support international students in maintaining meaningful contact with people who matter to them, helping to mitigate homesickness and loneliness.

Encourage participants to share their thoughts as they go. If they get stuck for ~30 seconds during a task, offer the hint shown. The goal is not to make them succeed — it is to observe how they try and how they reflect on the experience.

Introduction to read aloud

"Thank you for taking part. In a moment I'll show you a prototype of a mobile app called keepclose. The app is designed to help international students maintain regular contact with people who matter to them — family, close friends, anyone they care about. It is not a calling app itself, but a layer that sits alongside apps like WhatsApp and FaceTime to help keep in touch more regularly."

"I'll give you three short scenarios, one at a time. Please try to complete each one using the prototype on the screen. Feel free to comment as you go, ask questions, or share reactions. Quiet exploration is fine too. There are no right or wrong answers — if something feels confusing, that's useful for me to know. Ready?"

Scenario 1 — Drift to Call (Journey 1)

Prototype starts on Screen 1 (Contacts).

Scenario to read aloud

"Imagine you're an international student studying in the UK. Your mum lives back home. It's been a while since you last called her, and you've been meaning to. You open the app to see what the situation is, and maybe give her a call if it's a good time. Please show me what you'd do."

Key things to observe (brief)

- Did they notice the "Attention needed" framing?
- Did the time-zone info feel useful or ignorable?
- How did they react to the app picker appearing?

Hint (only if stuck)

"Take a look at the top of the contacts list. Is anyone flagged as needing attention?"

Post-task questions (open-ended)

Capture a paraphrase of their answer in the session notes. Mark anything quote-worthy with a star (★) for later expansion.

Q1. Imagine you'd been using this app for the last few weeks. How (if at all) would it have changed how often or how easily you'd call someone like Mum?

Q2. The app surfaces "Mum needs attention" rather than waiting for you to remember. How does that approach sit with you — would it help you keep in touch, or would it feel like another thing nudging you?

Q3. The app is designed as a layer on top of existing calling apps (e.g. WhatsApp, FaceTime, etc.), not as a replacement. Did that feel right to you, or would you want a different relationship between this and the calling apps you already use?

Scenario 2 — Add a Contact (Journey 2)

Prototype returned to Screen 1.

Scenario to read aloud

"Now imagine your sister Sara has just moved to Saudi Arabia for work. You want to add her to the app so you can stay in regular touch. Please show me how you'd add her as a new contact, and set up how often you'd like to call her."

Key things to observe (brief)

- Did the participant grasp the "Circle" idea naturally?
- How did they react to pre-filled defaults?
- Did they engage with the calling goal options or just accept the default?

Hint (only if stuck)

"Is there a way to add something new on this screen? Look around the edges."

Post-task questions (open-ended)

Q1. The app organises people into a "Home" circle (people back in your home country) and a "Local" circle (people in the country you're studying in). It also asks you how often you'd like to call each contact. Does thinking about contacts in this structured way feel useful for staying in touch, or does it feel artificial?

Q2. Setting a "calling goal" of e.g. once a week is a deliberate design choice — making the desired contact frequency explicit. Does that idea resonate with how you actually want to maintain relationships, or does it feel like the wrong way to think about staying in touch with people?

Scenario 3 — Review Patterns (Journey 3)

Prototype returned to Screen 1.

Scenario to read aloud
"It's the end of the month. You want to take a quick look at how regularly you've been in touch with people recently — just to get a sense of things. Please show me where you'd go and what you'd do."

Key things to observe (most important scenario — watch emotional reactions)

- How does the screen make them feel? Calm? Judged? Reassured?
- Did they understand the mini-strip pattern without explanation?
- Did they miss having any information — scores, streaks, totals?

Hint (only if stuck)
"Is there anywhere in the app you might look to see information over time?"

Post-task questions (open-ended)

Q1. How did looking at that screen make you feel?

Q2. The screen deliberately avoids scores, streaks, or "you missed X weeks" framing. Did the calmer approach work for you, or did you find yourself wanting more concrete feedback on how you're doing?

Q3. If you used this regularly, how do you imagine looking at this screen would influence (or not) your actual contact behaviour over the following weeks?

Overall questions (at end of session)

These are the most important questions in the session. Give participants time to think — don't rush them.

Q1. Stepping back from the specific features — do you think an app like this could actually help international students like yourself feel more connected to people who matter to them? Where might it succeed, and where might it fall short?

Q2. Is there anything you'd add or remove?

Closing the session

"That's the end. Thank you so much for your time".

Appendix C — Session-note template

Study: Supporting Homesickness and Loneliness in International Students — Designing a Wellbeing-Maintenance Calling Tool

Prototype: keepclose

One template per participant. Fill in during and immediately after the session.

Session details

Participant code: _____ **Date:** _____

Session length: _____ **Format:** _____

Format = in person / video call.

Participant background (light demographics)

Ask briefly at the start. No identifying info.

Time abroad: _____ **Home tz diff (hrs):** _____

Course / level: _____ **Phone OS:** _____

E.g. "First year BSc" / "MSc". Phone OS = iOS / Android.

Pre-session checklist

- Information sheet shared and read.
- Consent form completed.
- Verbal confirmation participant can stop / withdraw at any time.
- Prototype loaded on Screen 1.

Scenario 1 — Drift to Call

"Imagine you're an international student studying in the UK. Your mum lives back home. It's been a while since you last called her, and you've been meaning to. You open the app to see what the situation is, and maybe give her a call if it's a good time. Please show me what you'd do."

Task completion

Task / Step	Completed	Partial / hint	Failed
Found Mum in “Attention needed” section	☐	☐	☐
Tapped Mum's card to open detail screen	☐	☐	☐
Noticed and used the time-zone information	☐	☐	☐
Tapped Call now and saw the launch sheet	☐	☐	☐
Indicated they would launch a call	☐	☐	☐

Anything notable to flag

Post-task questions

Paraphrase responses in your own words. Use quote marks and a ★ for any phrases worth capturing verbatim — striking, surprising, or vivid wording that would work as a quote in the dissertation.

Q1. Imagine you'd been using this app for the last few weeks. How (if at all) would it have changed how often or how easily you'd call someone like Mum?

Q2. The app surfaces “Mum needs attention” rather than waiting for you to remember. How does that approach sit with you — would it help you keep in touch, or would it feel like another thing nudging you?

Q3. The app is designed as a layer on top of existing calling apps (e.g. WhatsApp, FaceTime, etc.), not as a replacement. Did that feel right to you, or would you want a different relationship between this and the calling apps you already use?

Scenario 2 — Add a Contact

"Now imagine your sister Sara has just moved to Saudi Arabia for work. You want to add her to the app so you can stay in regular touch. Please show me how you'd add her as a new contact, and set up how often you'd like to call her."

Task completion

Task / Step	Completed	Partial / hint	Failed
Identified and tapped the FAB (+ button)	☐	☐	☐
Reached the Add Contact form	☐	☐	☐
Tapped a field and reached filled state	☐	☐	☐
Tapped Save	☐	☐	☐
Reached final screen with Sara added	☐	☐	☐

Anything notable to flag

Post-task questions

Q1. The app organises people into a "Home" circle (people back in your home country) and a "Local" circle (people in the country you're studying in). It also asks you how often you'd like to call each contact. Does thinking about contacts in this structured way feel useful for staying in touch, or does it feel artificial?

Q2. Setting a "calling goal" of e.g. once a week is a deliberate design choice — making the desired contact frequency explicit. Does that idea resonate with how you actually want to maintain relationships, or does it feel like the wrong way to think about staying in touch with people?

Scenario 3 — Review Patterns

"It's the end of the month. You want to take a quick look at how regularly you've been in touch with people recently — just to get a sense of things. Please show me where you'd go and what you'd do."

Task completion

Task / Step	Completed	Partial / hint	Failed
Found the History tab in bottom nav	☐	☐	☐
Reached the History screen	☐	☐	☐
Understood the mini-strip pattern	☐	☐	☐
Switched between 4-week / 8-week ranges	☐	☐	☐
Indicated they had finished reviewing	☐	☐	☐

Anything notable to flag (most important scenario — watch emotional reactions)

Post-task questions

Q1. How did looking at that screen make you feel?

Q2. The screen deliberately avoids scores, streaks, or "you missed X weeks" framing. Did the calmer approach work for you, or did you find yourself wanting more concrete feedback on how you're doing?

Q3. If you used this regularly, how do you imagine looking at this screen would influence (or not) your actual contact behaviour over the following weeks?

Overall (end of session)

Most important questions in the session — give participant time to think.

Q1. Stepping back from the specific features — do you think an app like this could actually help international students like yourself feel more connected to people who matter to them? Where might it succeed, and where might it fall short?

Q2. Is there anything you'd add or remove?

Researcher reflection (fill in immediately after session)

Your gut impression while fresh — used during analysis.

Top 2–3 takeaways from this session:

Anything that surprised you (positively or negatively):

Tags / themes (keywords — e.g. "would help me call more", "nudge felt pushy", "wants a streak"):

End of notes for participant code: _____

Save as: P0X_session_notes.docx (e.g. P01_session_notes.docx)

Appendix D — Participant info sheet

Study: Supporting Homesickness and Loneliness in International Students — Designing a Wellbeing-Maintenance Calling Tool

Researcher: Ayoub Aldawood, BSc Computer Science (Software Engineering), Newcastle University

Supervisor: Nick Taylor

Module: CSC3094 Major Project and Dissertation in Computer Science.

Invitation to take part

You are being invited to take part in a short research study about a mobile app prototype designed to help international students maintain regular contact with people who matter to them. Before you decide whether to take part, it is important that you understand what the study involves and what you will be asked to do. Please read this information sheet carefully and ask any questions you have.

What is the purpose of this study?

International students frequently describe homesickness and loneliness as significant parts of their experience abroad. Staying in regular contact with family and close friends is one way students cope with being far from home — but everyday pressures make this harder than it sounds. This project has designed a mobile app prototype that aims to help international students maintain these connections through routine-based, low-pressure calling.

This evaluation session will help me understand whether the prototype is easy to use, whether it supports the kinds of calling behaviours students actually want, and how it makes people feel. Your feedback will directly inform the prototype and the dissertation that accompanies it.

Why have I been asked to take part?

You have been invited because you are an international student, which makes you part of the target user group for this app. You do not need any technical background or design experience — I am interested in your honest reactions as a potential user.

What will I be asked to do?

The session will last approximately 25 to 30 minutes and will take place in person. You will be asked to:

- Complete three short tasks using an interactive prototype of the app on a phone or laptop screen.
- Feel free to comment as you go — say what you notice, ask questions, or share reactions if you want to. Quiet exploration is fine too.
- Answer a short set of questions after each task about your experience.
- Answer two general questions at the end about your overall impression of the app.

There are no right or wrong answers. The prototype is being evaluated, not you. If something feels confusing or frustrating, that is valuable information.

Do I have to take part?

No. Participation is entirely voluntary. You can withdraw at any point during the session without giving a reason, and you can ask for your data to be.

Will my data be kept confidential?

Yes. Your participation is anonymous. I will not collect your name, email address, or any information that could identify you personally. You will be assigned a participant code (e.g. P01, P02) and all notes and responses will be stored under that code only.

No audio or video recording will take place unless you explicitly consent to it.

How will my data be used?

Your responses and observations from the session will be analysed alongside those of other participants to identify patterns, strengths, and weaknesses of the prototype. Findings will be reported within my CSC3094 dissertation. Direct quotes may be used in the dissertation but will never be attributed to you by name or any identifying detail.

What are the possible risks of taking part?

This study is low risk. You will not be asked to share sensitive personal information about your own relationships, mental health, or experiences of homesickness. The tasks involve interacting with a prototype in hypothetical scenarios. If any question or topic feels uncomfortable, you are free to skip it or end the session.

What are the possible benefits of taking part?

There is no direct benefit or payment for taking part. However, your participation will contribute to my work that aims to improve the wellbeing of international students, and you will have had the opportunity to try a prototype and share your views on it.

Who has reviewed this study?

This study has been conducted in line with Newcastle University's ethics guidance for undergraduate student projects and has been discussed with my project supervisor.

What if I have questions or concerns?

You are welcome to ask me any questions before, during, or after the session. If you have concerns that you do not feel comfortable raising with me directly, you can contact my project supervisor.

Researcher contact: Ayoub Aldawood — A.N.A.Aldawood2@newcastle.ac.uk

Supervisor contact: Nick Taylor — nick.taylor@newcastle.ac.uk

Thank you for considering taking part in this study.

Appendix E — Consent form

Study: Supporting Homesickness and Loneliness in International Students — Designing a Wellbeing-Maintenance Calling Tool

Researcher: Ayoub Aldawood, Newcastle University, CSC3094 Major Project and Dissertation.

Please read each statement below and tick the Yes or No box to indicate whether you agree.

Statement	Yes	No
I confirm that I have read and understood the participant information sheet for this study.	☐	☐
I have had the opportunity to ask questions about the study and have received satisfactory answers.	☐	☐
I understand that my participation is voluntary, and that I may withdraw at any time during the session without giving a reason.	☐	☐
I understand that I may request deletion of my data.	☐	☐
I understand that no identifying information (name, email, or personal details) will be collected, and that my responses will be stored under a participant code.	☐	☐
I understand that direct quotes from my responses may be used in the dissertation, but that they will not be attributed to me.	☐	☐
I consent to the researcher taking written notes during the session.	☐	☐
I agree to take part in this study.	☐	☐

Participant code: _____

Date: _____

Thank you for taking part in this study.